

LESSON 1: INTRODUCING COMPUTERS

Computers are all around us. We use them in almost every aspect of our lives. But what are they? How do we best define them? What do they do for us?

A **COMPUTER** IS A TECHNOLOGY WHOSE PURPOSE IS TO PROCESS INFORMATION ACCORDING TO A STORED PROGRAM.

TECHNOLOGY

The word technology refers to the tools and machines we use to help us perform various tasks. Every technology has a specific purpose; it is designed for a particular task.

EXAMPLE 1.1

A pencil is a form of technology. It is a tool designed to help us perform the task of writing. Even though there are many different types of pencils, they are all designed to perform the same task, writing.



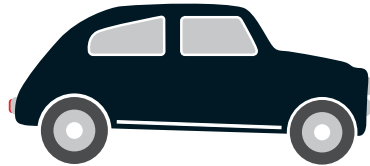
Notes

The term 'technology' encompasses more than simply tools and machines; it includes any practical application of knowledge.

The Merriam-Webster Dictionary defines technology as "the practical application of knowledge, especially in a particular area."

EXAMPLE 1.2

A car is also a form of technology. It is a machine designed to help us travel from one place to another on land. Regardless of the shape, size and color, all cars are designed for the specific task of transporting people from one place to another on land.



EVERY TECHNOLOGY IS DESIGNED FOR A PURPOSE.

PROCESSING

To process something means to change it from one form to another. Processing is not any kind of change; it is a very structured form of change that can be replicated. A process is structured according to a specific set of steps; if given the same starting conditions and materials, these steps will provide the same end result every time they are performed.

EXAMPLE 1.3

Processed food is food that undergoes change starting from certain raw materials. Bread is processed food. It starts as wheat, a raw material that is unprocessed. The processing of wheat to make bread must follow a specific set of steps.

A PROCESS IS A SERIES OF STEPS THAT COMPLETE A PARTICULAR TASK AND CAN BE REPLICATED.

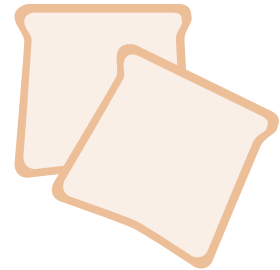
Anyone who replicates these steps will make bread from wheat. Regardless of how many times the steps are repeated and regardless of who performs these steps, the processing of wheat will always result in bread.



PROCESSING OF WHEAT TO BREAD

STEPS

1. *Clean the wheat grains*
2. *Grind the wheat grains*
3. *Sift the flour to remove the husk*
4. *Send the flour to the bakery*
5. *Use flour to make bread dough*
6. *Bake the dough*



INFORMATION

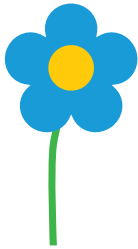
In general, the term information refers to anything meaningful that tells us something about ourselves or the world around us. This book contains information. The information is conveyed in the words and pictures of this book.

As humans, we process information. We see, feel and hear things that we process to gain information about the world around us. We retain this information in our memories. Later, we may use this information to interact with the world.

EXAMPLE 1.4

Sarah and her friends were playing the “Guess the Object” game. Sarah was blindfolded, and her friends laid out objects in front of her. Using her senses, she had to gather information about these objects to guess what they were.

Even without seeing the objects, Sarah could identify them. In addition to each object's shape, which she could feel, the flowers smelled nice, the keys jingled and the ice cube felt cold.



Sarah's senses helped her gather information about each object. But that was not enough to identify each object. Sarah must have had in her memory some information about the characteristics of each object against which she could compare. The new information gathered by her senses was compared with the information she had in her memory.



[1]

Sarah gets information by smelling the object.



[2]

Sarah processes the information by comparing it with previous information in her memory.

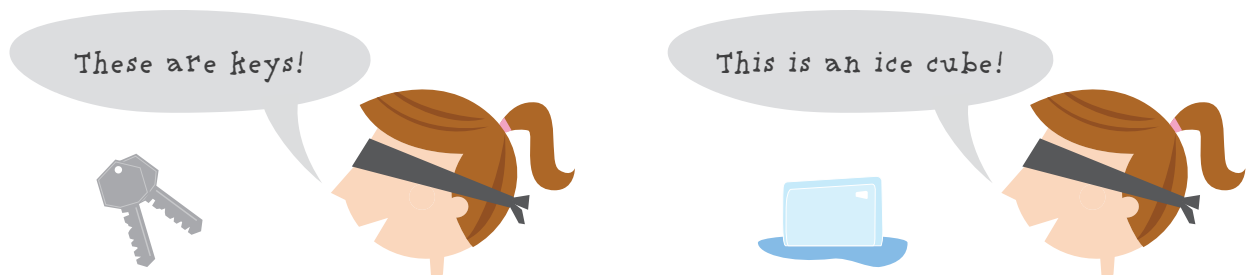
To identify the object, Sarah's brain had to process information. Sarah's brain had to go through a series of steps in which she compared the information acquired through her senses with the information she had in her memory. By making these comparisons, Sarah was able to identify the object and state its name. The information acquired through her senses was processed; it was changed to the name of the object.

[3]

Sarah informs her friends that the object is a flower.



Sarah followed the same steps to identify the rest of the objects. Her senses acquired information about each object. She compared that information with that in her memory. An object was identified when a match was found. She would then inform her friends of the name of each object.



[4]

Sarah follows the same steps to identify the other two objects.

COMPUTERS

Computers are a form of technology designed for the purpose of processing information. This includes creating information, sharing information and storing information, all of which require some form of processing.

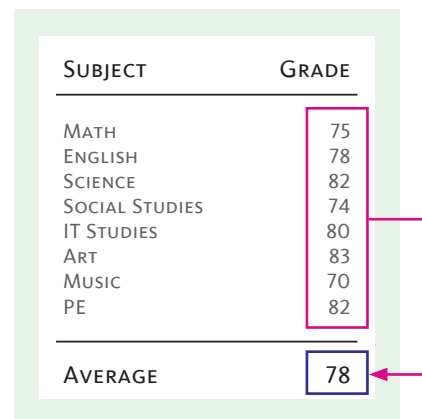
CREATING INFORMATION

People use computers to create new information. When we type a text message, email or essay, in each case, we are creating new information. When we record a voice message or video, we are creating new information. In this context, information is considered to be any meaningful content. Making new content, whether in the form of text, audio or images, is all made possible by computers.

Creating new information is not limited to typing an essay, recording a voice message or capturing an image. Creating new information also includes the processing of data either through calculation or analysis.

EXAMPLE 1.5

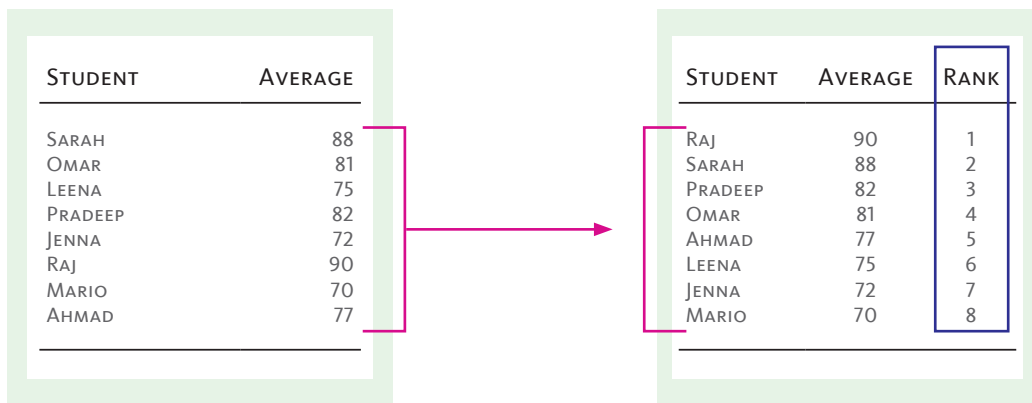
Every school keeps track of its students' grades. At the end of each term, the school issues a report card to every student. The computer helps create every report card. Teachers enter the grades for each subject into the computer system. The computer then processes this information to provide an average for each student. The average is the new information.



SUBJECT	GRADE
MATH	75
ENGLISH	78
SCIENCE	82
SOCIAL STUDIES	74
IT STUDIES	80
ART	83
MUSIC	70
PE	82
AVERAGE	78

Data is processed to calculate the average.

The school ranks its students. To do so, the list of students needs to be sorted from the highest average to the lowest. The computer processed this information and sorted the list of students. This provides the school with the student ranks, which is new information.



Data is processed to sort the list of students from highest to lowest average. The rank is new information.

SHARING INFORMATION

Creating information and new content becomes more valuable when it can be shared with others. Computers have tremendously increased our ability to share information.

Sharing information is made possible by the Internet, a complex network of telecommunication technologies that spans the globe and allows computers to connect to each other and to other digital devices.

THE INTERNET IS A COMPLEX NETWORK OF TELECOMMUNICATIONS TECHNOLOGIES THAT CONNECTS COMPUTERS ACROSS THE WORLD.

All information accessible through the Internet is referred to as the World Wide Web or simply as the Web.

Information on the Web is organized into files. The content of each file can be text, audio or images. This content is displayed as webpages, music, pictures or video. Each file has a specific address, a web address, that is used to locate it.

AN ADDRESS IS USED TO LOCATE INFORMATION.

Information on the Web is shared as either private or public, which determines who gains access to this information. Access to private information is restricted and requires a password. In contrast, public information does not require a password and is accessible to everyone.

EXAMPLE 1.6

Your school may have a website. The school website is made up of several web pages that contain information made public by your school. In addition, the school may offer its students a portal through which they can sign in with their passwords to access their private information, such as their grades.

USERNAME:

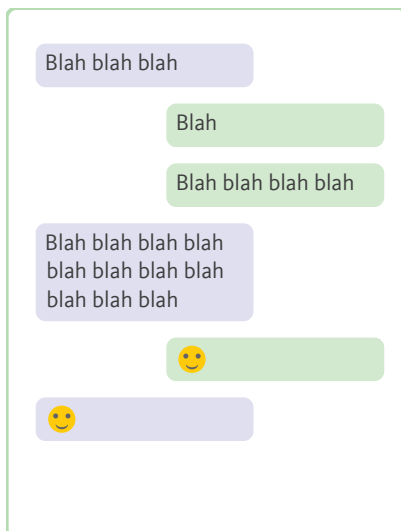
PASSWORD:

COMMUNICATION IS A FORM OF SHARING INFORMATION THAT RESULTS IN AN EXCHANGE BETWEEN TWO OR MORE PEOPLE.

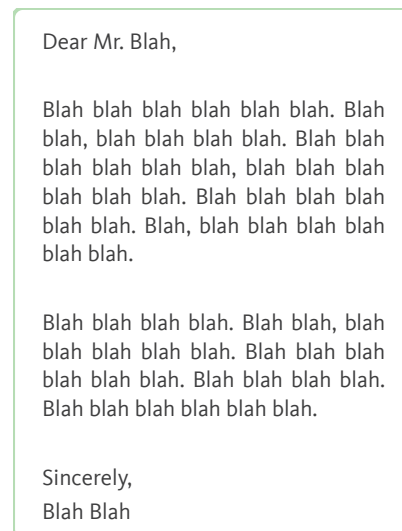
Another form of sharing information is communication. Communication is sharing that results in an exchange of information between two or more people.

Two methods of communication facilitated by computers are messaging and emailing. Both methods allow for the exchange of information in the form of text and audiovisual content. Messaging is better suited for shorter and more frequent exchanges of messages, whereas emailing is better suited for more prolonged and less frequent exchanges of messages.

Text messages



Email



Messaging and emailing are two forms of communication facilitated by computers.

STORING INFORMATION

Once information has been created, it must be stored. Storage of information is important. If information were volatile, meaning it “evaporates” whenever a computer was powered off, no information would be left. All our information would be lost; every message, every essay and every photo would be lost.

Every computer must be capable of storing information. Storage provides a permanent repository for information. Computers not only allow us to store our information, but they also provide us with the facilities to search and find specific items of information kept in storage.

STORAGE IS THE PERMANENT REPOSITORY FOR INFORMATION.

All information is stored in the form of files. Every file is given a filename by which it can be identified. Every file is also given a specific address at which it can be located. The size of a file is highly variable; it can contain 1 byte or several gigabytes of information. The content of a file may be mostly instructions, which is the case for apps, or it may only be data, as is the case for photos.

A FILE IS A UNIT OF STORABLE INFORMATION.

THE STORED PROGRAM

The stored program concept is an important concept that made the modern computer possible. Imagine your computer without any apps. What could it

possibly do? Would it be of any use to you?

The fact is that every app on your computer performs a particular task. If you want to compose an essay, you use a word processing app. If you want to record a video, you use a camera app. If you want to search for information on the Web, you use a web browser app. Each of these apps is a program stored in your computer, and each program contains the instructions the computer must follow to perform that particular task.

A PROGRAM IS A SET OF INSTRUCTIONS INTENDED TO PERFORM A PARTICULAR TASK.

In addition to the apps you use, there are other stored programs on your computer that you do not see and do not directly use, but they are critical for making your computer work the way it does. Most of these programs are part of the operating system, which is a group of programs that work together to allow your computer to use a variety of apps and to function properly.

SUMMARY

The word technology refers to the tools and machines we use to help us perform various tasks. Every technology is designed for a purpose.

Processing is a structured form of change that can be replicated. Computers are designed for the purpose of processing information.

Computers allow us to create, share and store information.

The Internet is a complex network of telecommunications technologies that connects computers across the world. It is used to share and exchange information. Access to shared information is either restricted, private or unrestricted, public.

An address is used to locate information. A web address is an address used to locate information on the World Wide Web.

Communication is a form of sharing information that results in an exchange between two or more people. Emailing and messaging are two examples of methods of communication.

Storage is the permanent repository for information. All information is stored as files. A file is a unit of storable information. Every file has a filename by which it is identified and an address by which it is located.

A program is a set of instructions intended to perform a particular task. Stored programs allow the computer to perform different tasks. An app is an example of a stored program.

A computer is a technology whose purpose is to process information according to a stored program.

Activity 1.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>T</u>	Information is any meaningful content.
<u>F</u>	Computers are not a form of technology.
<u>T</u>	Every technology has a purpose.
<u>T</u>	Processing generally means changing something in accordance with a series of steps.
<u>F</u>	Information cannot be processed.
<u>F</u>	Computers allow us to only share information.
<u>T</u>	Storage is the permanent repository for information.
<u>T</u>	A program is a set of instructions intended to perform a particular task.
<u>F</u>	An application is not an example of a stored program.
<u>T</u>	Stored programs made modern day computers possible.

Activity 1.2

Identify machines, appliances and devices for each of the forms of technology based on its stated purpose.

1. Which two of these machines are forms of technology designed for the purpose of transportation?
 - a) ship
 - b) microwave oven
 - c) airplane
 - d) computer

2. Which two of these appliances are forms of technology designed for the purpose of cooling food?
 - a) washing machine
 - b) freezer
 - c) air conditioner
 - d) refrigerator

3. Which two of these devices are forms of technology designed for the purpose of generating electrical energy?
 - a) wind turbine
 - b) phone battery
 - c) solar panel
 - d) electric motor

4. Which two of these devices are forms of technology designed for the purpose of processing information?
 - a) laptop
 - b) television
 - c) printer
 - d) desktop

Activity 1.3

Answer the questions below.

1. Transportation technologies can be further categorized based on the medium within which they are intended to travel. List two machines designed for the purpose of traveling in each medium.

Land : car, train, bus

Air : airplane, helicopter, hot-air balloon

Water : ship, submarine, yacht

2. Investigate the two forms of technology. Identify and state the purpose of each.

Nanotechnology : its purpose is to manipulate matter at the tiniest scale, the nanoscale (1 -100 nm), to make new products.

Biotechnology : its purpose is to use biological processes, living cells and organisms to make new products.

3. “Recognizing the purpose of any particular form of technology is important.” Analyze this statement, investigate and provide at least two supporting reasons.

Recognizing the purpose behind a particular technology allows us to:

1) understand how to use it.

2) determine if it will be useful to us.

3) avoid potential risks.

Activity 1.4

Answer the questions below.

1. What information do drivers get when they see a red traffic light?
 - a) go
 - b) **stop**
 - c) turn left
 - d) turn right

2. What information do drivers get when they hear the sirens of an ambulance behind them?
 - a) **move out of the way**
 - b) block the road
 - c) get help
 - d) cover your ears

3. If you were blindfolded and asked to guess if an object in front of you was an ice cube, what information would you be searching for?
 - a) **cold, wet, smooth**
 - b) warm, dry, rough
 - c) cold, wet, rough
 - d) warm, wet, smooth

4. For each of the questions above, state the sense through which information was acquired.
Question 1: **sight**
Question 2: **hearing**
Question 3: **touch**

Activity 1.5

Answer the questions below.

1. Which of the following best describes a process?
 - a) A random sequence of events
 - b) A type of machine used in manufacturing
 - c) A clear sequence of steps for completing a task
 - d) A finished product

2. Which of the following is a characteristic of a well-defined process?
 - a) It is constantly changing
 - b) It is repeatable and consistent
 - c) It does not have a clear beginning
 - d) It does not have clear steps to follow

3. Which of the following is not an example of a process?
 - a) Baking a cake
 - b) Getting dressed
 - c) Watching a movie
 - d) Brushing teeth

4. Why do you think it is important to understand a process?
 - a) It helps us complete the task quickly and properly
 - b) It allows us to skip steps in the process
 - c) It makes the process more complicated than it needs to be
 - d) It ensures that you don't need to follow any specific steps

Activity 1.6

Answer the questions below.

1. What information do you need to bake a cake?
 - a) Which ingredients go into making a cake
 - b) What steps need to be performed
 - c) Which utensils are required to perform the steps
 - d) All the above

2. What item of information would be useful for you to navigate your way out of your classroom?
 - a) The color of the walls
 - b) The location of the exit
 - c) The names of the students
 - d) The number of open windows

3. “Information is critical to the process of decision-making.” Is this statement true? Why? Give an example.

Yes, this statement is true. Having the correct information allows us to make good decisions and to avoid making mistakes. The lack of information or having the wrong information can lead to unwanted outcomes. For example, when going on a road trip, having the wrong location for the destination or not having the location may result in us getting lost and not arriving at the destination.

Activity 1.7

Match each word with the statement that best describes it.

	File	Filename	Program	Address
	Web Address	Internet	Storage	
1.	A telecommunications technology that spans the globe	Internet		
2.	A unit of storable information	File		
3.	It is used to locate a file	Address		
4.	It is used to identify a file	Filename		
5.	It is used to locate a file across the Internet	Web Address		
6.	A computer's permanent repository of information	Storage		
7.	A set of instructions intended to perform a task	Program		

Activity 1.8

For each of the following scenarios, identify how information was created, stored and shared.

- Karl watches a video on YouTube.

Created : The video is the information created by someone.

Stored : The video is stored on a YouTube computer (server).

Shared : The video is viewed by people who find it , like Karl.

2. Jenna uses her smartphone to send a voice message to her friend.

Created : The voice message is the information created.

Stored : The voice message is stored on Jenna's smartphone.

Shared : The voice message is sent through the Internet to her
friend's phone.

3. A teacher enters her students' grades. Later, parents view their children's report cards which contain their general average and rank.

Created : 1) Grades are entered. 2) General averages are calculated.
3) Ranks are identified by sorting based on general averages.

Stored : Student grades, general averages and ranks are stored in
the school's main computer (server).

Shared : Parents access the school's main computer using the Internet
and view their child's grades, general average and rank.

4. Describe your own example of using a computer to create, store and share information.

Activity 1.9

Match each word with the statement that best describes it.

	Program	Private	Messaging	Public
	Email	Applications	Images	
1. Frequent and short exchanges of information			<u>Messaging</u>	
2. Information that has restricted access			<u>Private</u>	
3. Information that has unrestricted access			<u>Public</u>	
4. Communication suitable for long, infrequent exchanges	<u>Email</u>			
5. Examples of files that contain only data			<u>Images</u>	
6. Examples of files that contain both instructions and data		<u>Applications</u>		
7. A set of instructions that perform a particular task		<u>Program</u>		

Activity 1.10

Answer the questions below.

- Two common methods of communication are messaging and email. Describe how the two methods differ.

Messaging may be more effective for quick and informal communication with
friends and classmates, while email may be more effective for more formal or
professional communication with teachers or business partners.

2. In the previous question, the differences between messaging and email were described. Describe how the two methods are similar.

1) Both are forms of digital communication.

2) Both require the Internet to transfer information.

3) Both can be used to send information in the form of text, audio and video.

3. Every application performs a particular task. List two applications and what you use them for.

Answers may vary. For example, I use PowerPoint to prepare a presentation

for my classmates about my science project.

4. List at least three ways you will use computers to help you learn.

Possible answers may include reference to: 1) access to educational

resources, 2) collaboration and communication, 3) interactive learning,

4) personalized learning and 5) research and data analysis.

LESSON 2: TYPES OF COMPUTERS

There are many types of computers. In general, computers can be classified based on their intended purpose into two broad categories: general-purpose and specific-purpose.

COMPUTERS CAN BE CLASSIFIED BASED ON THEIR PURPOSE AS
GENERAL-PURPOSE OR SPECIFIC-PURPOSE.

GENERAL PURPOSE

General-purpose computers are the most common and widely used computers. They are our personal computers: the desktop, laptop, tablet and smart-phone. These computers are designed to run a wide variety of applications which allows them to perform a wide variety of tasks that depend on the needs of the user.

So rather than being designed to perform one specific task or set of tasks, these general-purpose computers are designed to be highly flexible and to perform a wide variety of tasks. The choice of the task depends on the user and the availability of an application to perform the required task.

PERSONAL COMPUTERS ARE GENERAL-PURPOSE COMPUTERS.

EXAMPLE 2.1

Karl has a laptop that he uses for many different purposes. Here is a list of some of the tasks he can perform using the various applications he has on his laptop.

- **Writing assignments:** Using a word processing application, such as MS Word, Karl can type essays, reports or other school projects.
- **Multimedia presentations:** Using applications, such as MS PowerPoint, Karl can prepare slideshows with images and videos for his school projects.
- **Research:** Using a web browser application, such as Google Chrome, Karl can find information in the forms of articles, e-books, images and videos.
- **Collaboration:** Using applications, such as Google Drive, Karl can share his projects with his classmates so that they can complete them as a team.
- **Video Conferencing:** Using applications, such as Google Meet, Karl can communicate with his classmates to enhance collaboration on projects.
- **Gaming:** Karl has a variety of games that he can play alone or with his friends online, such as Minecraft and Fortnite.
- **Entertainment:** Using applications, such as Netflix, Karl can watch movies for entertainment.
- **Time Management:** Using applications, such as Calendar, Karl can schedule his assignments and ensure they are completed on time.

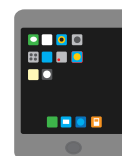
These are some of the many tasks for which Karl uses his computer. The general-purpose nature of his laptop allows him to adapt it to his various needs and preferences, which makes it a versatile and useful machine.



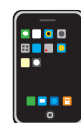
Desktop



Laptop



Tablet



Smartphone

PERSONAL COMPUTERS

Personal computers are classified into four types: desktop, laptop, tablet and smartphone computers. Flexibility is a feature common to all personal computers. Two features that distinguish these computers are portability and capability.

**PERSONAL COMPUTERS CAN BE DISTINGUISHED BASED ON
PORTABILITY AND CAPABILITY.**

Portability refers to the ease with which a computer can be carried around. This depends on three factors: size, weight and power supply. Certainly, the size and weight of a computer affect the ease with which it is carried around, making smaller and lighter computers more portable than larger and heavier ones. In addition, having a portable electrical power source, a battery, makes portability possible. Computers that need to remain connected to a power outlet remain stationary and are not portable.

**PORTABILITY DEPENDS ON THE SIZE, WEIGHT AND POWER SOURCE
OF THE COMPUTER.**

Capability refers to the complexity of tasks a computer is capable of performing. Complex tasks typically require a larger computer with a larger screen and, more importantly, greater computational power. The larger screen allows the user to clearly see the various visual elements displayed made possible by the greater computational power which allows the computer to process a lot of information and to perform many complex calculations very fast. However, computational power requires electrical power. The more information a computer processes per second, the more electrical energy it requires per second.

CAPABILITY DEPENDS MAINLY ON COMPUTATIONAL POWER.

1. **Desktops** are large, stationary computers that are designed to perform more complex tasks. Therefore, they have larger screens and greater computational power, both of which demand more electrical energy. This makes the battery an inconvenient source of electrical power as it will require frequent recharging. In addition, since complex tasks are typically performed with the user seated at a desk, a battery is not required.
2. **Laptops** are portable computers with capabilities close to those of a desktop. To achieve portability, laptops are made smaller by combining the screen and keyboard into one unit rather than being separate parts, as in desktops. In addition, laptops are powered by a rechargeable battery which can handle the energy demands of the smaller screen and the less complex tasks. Do note that certain laptops can match the capabilities of some desktops.
3. **Tablets** are not only portable; they are handheld devices that are larger than smartphones but smaller than laptops. Their smaller size is achieved through touchscreen technology which abates the need for a physical keyboard and pointing devices, such as the mouse and touchpad. They are more portable but less powerful and, therefore, less energy-demanding than laptops.
4. **Smartphones** are portable handheld devices designed primarily for communication but can also be used for a wide range of other tasks, such as browsing the internet, taking pictures and playing games. They are the most portable of the personal computers and the least capable.

In general, personal computers are designed to be usable by a wide range of people and in a wide range of settings. People of all ages, backgrounds and professions find the personal computer very useful in their lives. So not only are personal computers designed to be highly flexible, highly usable and with decent computational power, but they are also designed to deliver all that at an affordable price.

PERSONAL COMPUTERS ARE GENERALLY DESIGNED TO BALANCE CAPABILITY, FLEXIBILITY AND USABILITY WITH AFFORDABILITY.

SPECIFIC PURPOSE

In contrast to general-purpose computers, specific-purpose computers are designed to perform a specific set of functions or tasks. These computers are often specialized and designed to run specific applications rather than a broad range of applications.

SUPERCOMPUTERS

The supercomputer is a type of computer that is designed to solve complex problems. Typically, a supercomputer will handle one complex task at a time. Such a task requires massive amounts of computational power in order to handle the massive amounts of information that must be processed. Therefore, supercomputers are currently massive in size and demand immense amounts of electrical energy, making them very expensive to have and maintain.

Usually, these computers are used by major institutions, corporations and governments for the purposes of scientific research, weather forecasting, oil and gas exploration, molecular modeling and other large-scale applications.

SUPERCOMPUTERS ARE DESIGNED TO HANDLE ONE COMPLEX TASK AT A TIME.



A supercomputer focuses all its computational power on addressing one big task.

EXAMPLE 2.2

The IBM Blue Gene used by the National Oceanic and Atmospheric Administration (NOAA) in the United States is a supercomputer used for weather forecasting. The Blue Gene can perform over 1 quadrillion calculations per second, making it one of the most powerful computers in the world.

SUPERCOMPUTERS ARE OPTIMIZED FOR PERFORMANCE.

The Blue Gene can analyze vast amounts of meteorological data, including satellite images, weather station readings and oceanic data, in order to create detailed models of the atmosphere. These models can then be used to forecast weather patterns, including the trajectory of storms, the likelihood of severe weather events and long-term climate trends.

This allows for more accurate and timely predictions of severe weather events, which has important implications for disaster preparedness and response, as well as for industries such as agriculture and transportation that are heavily impacted by weather patterns.



The IBM Blue Gene/P supercomputer installation at the Argonne Leadership Computing Facility located in the Argonne National Laboratory in Lemont, Illinois, USA

SERVERS

The server is a computer that is designed to handle many simple tasks at the same time, rather than one complex task at a time, as would the supercomputer. The majority of these simple tasks are requests for sharing information, such as videos on the YouTube server or for processing information, such as the calculation of student averages in the database of the school server.

SERVERS ARE DESIGNED TO HANDLE MULTIPLE TASKS CONCURRENTLY.



A server focuses all its computational power on handling as many requests as it can simultaneously.

Servers vary in size depending on the workload they have to handle. Servers that are expected to receive heavy workloads are combined together, forming a data center. The combined servers are large enough to have the required computational power but require a lot of energy and are expensive. On the other hand, servers that are expected to receive light workloads tend to be smaller in size, have less computational power, require less energy and are far less expensive. Servers, therefore, vary from data centers the size of a supercomputer to a single server computer the size of a desktop.

EXAMPLE 2.3

The number of requests received by a Google data center, which houses its servers, varies depending on a range of factors, such as the time of day, the location of the data center and the specific services being used. However, it is estimated that Google processes billions of search queries each day, translating to tens of thousands of requests per second across its global network of data centers. In general, each data center may handle millions or even billions of requests per day, depending on the demand for Google's services in that particular region.



One of Google's many data centers is located in St. Ghislain, Belgium. The servers in the data center run Google products such as Search, Gmail and YouTube, as well as provide support for Google users across Europe and around the world.

EXAMPLE 2.4

Jenna's school uses a server located on-site to store and share educational resources. Jenna's teachers upload course materials, assignments and student grades to the server, which students like Jenna can access from computers on the school's network. The server also provides Jenna's parents access to her grades, attendance records and other important information about her academic progress.

Jenna's school is a medium-sized school attended by a few hundred students. For a school that size, its server would likely handle a few hundred to, at most, a few thousand requests per day. This number depends on the demand placed on the server by students, teachers and parents.



The server at Jenna's school can handle a few hundred to a few thousand requests per day, depending on the demand placed on it by the students, teachers and parents.

Servers are classified as specific-purpose devices because they are designed to perform a specific set of functions or services. Servers are labeled according to the type of function they are dedicated to performing.

Web servers	Host websites and deliver web pages
Email servers	Manage email accounts and deliver email
File servers	Store and share files
Database servers	Manage and store data in database
Game servers	Host online games and manage player accounts

Servers are dedicated to serving other computers and their users. In general, most users expect servers to respond quickly, be available any time they send a request and, most importantly, to keep their private information secure. This means that servers are designed for high performance, reliability and security.

SERVERS ARE OPTIMIZED FOR PERFORMANCE, RELIABILITY AND SECURITY.

EMBEDDED COMPUTERS

Embedded computers are computers within other devices or systems, such as cars, medical devices and home appliances and are typically hidden from view. They are classified as specific-purpose because they are designed to perform specific tasks, particularly controlling and optimizing the function of the machine in which they are embedded. In general, embedded computers are optimized for energy efficiency, small size and high reliability.

EMBEDDED COMPUTERS ARE OPTIMIZED FOR LOW POWER CONSUMPTION, SMALL SIZE AND HIGH RELIABILITY.

1. **Low power consumption:** Embedded computers are often designed to operate in environments where power may be limited, such as in remote areas or in devices that rely on battery power. To conserve power and extend battery life, embedded computers are typically designed to be energy efficient and have low power consumption.

2. **Small size:** Embedded computers are often used in small devices or appliances with limited space. This means that they must be designed to use less space and have small components that can be densely packed.
3. **High reliability:** Embedded computers are often used in devices where reliability is critical, such as in automotive or medical devices. They must be designed to operate without interruption or failure, even in harsh environments or extreme conditions. This requires components that can withstand high temperatures, shock, vibration and other stresses.

EXAMPLE 2.5

Jenna's father has a heart condition. His cardiologist informed him that his condition requires a pacemaker. The pacemaker is a device that is placed inside the body and is connected to the heart; its task is to regulate the beating of the heart.

The pacemaker has an embedded computer. The computer is tasked with reading the electrical activity of the heart and sending regular electrical pulses to maintain a steady rhythm.

Low power consumption

Since the pacemaker is placed inside the body, it must be powered by a battery. Recharging the battery requires a trip to the doctor's office, so having a long battery life is convenient.

Small size

Since the pacemaker has to fit inside the body, it should be small in size. This means its embedded computer should also be small in size.

High reliability

Since the pacemaker is responsible for keeping Jenna's father alive, it must be reliable. It must not fail. If it fails, his life will be at risk.

Overall, embedded computers are designed to meet the specific needs and enhance the functionality of the device or machine they are being used in.

SUMMARY

Computers can be classified into two broad categories: general-purpose and specific-purpose.

General-purpose computers are flexible and can run a wide range of applications and therefore perform various tasks and serve many purposes. Personal computers fall into the category of general-purpose; they are: desktops, laptops, tablets and smartphones.

Personal computers are distinguished by portability and capability. Portability depends on the size, weight and power source. Capability is largely determined by computational power. The desktop ranks the lowest in portability but the highest in capability. The smartphone ranks the highest in portability but the lowest in capability, with the laptop and tablet ranking in between. Personal computers are designed to have reasonable capability, high flexibility, easy portability and affordability.

Specific-purpose computers are designed to perform a specific set of functions or tasks. Supercomputers, servers and embedded computers are types of specific-purpose computers.

Supercomputers are massive computers that are optimized for high performance regardless of the energy consumption and the high expenses required to own, run and maintain them. They handle one complex task at a time.

Servers vary in size from data centers as large as a supercomputer to a single server computer the size of a desktop, depending on the workload they are required to handle. In all cases, these computers are optimized for high performance, high reliability and high security.

Embedded computers are computers placed inside machines and devices for specific purposes that optimize their overall function. Embedded computers are optimized for energy efficiency, small size and high reliability. They are ideal for use in devices where space and power are limited and reliability is critical.

Activity 2.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>F</u>	General-purpose computers are designed to perform specific tasks.
<u>T</u>	Desktop computers are generally more powerful than laptops.
<u>T</u>	Size, weight and power supply are the three main factors that determine the portability of a computer.
<u>F</u>	Supercomputers can handle multiple complex tasks at the same time.
<u>F</u>	Servers can handle only one task at a time.
<u>F</u>	Portability is an important feature of servers.
<u>T</u>	Embedded computers are designed to be part of other devices.
<u>T</u>	Servers are typically more powerful than personal computers.
<u>T</u>	Embedded computers are optimized to be energy efficient, small in size and highly reliable.
<u>T</u>	There are different types of servers, each dedicated to a particular task.

Activity 2.2

Answer the questions below.

1. Which type of computer is designed to perform a wide range of tasks?
 - a) Specific-purpose computer
 - b) **General-purpose computer**
 - c) Embedded computer

2. Which of the following is an example of a specific-purpose computer?
 - a) Laptop
 - b) Desktop
 - c) **Supercomputer**

3. What is the primary function of a server?
 - a) To perform a wide range of tasks
 - b) **To provide resources or services to other computers on a network**
 - c) To control the engine of a car

4. What is the primary function of an embedded computer?
 - a) To perform a wide range of tasks
 - b) To provide resources or services to other computers on a network
 - c) **To perform a specific set of functions within a larger device or system**

5. Which type of computer is designed for portability and convenience?
 - a) Desktop
 - b) **Laptop**
 - c) Server

Activity 2.3

Answer the questions below.

1. Which type of computer is optimized for advanced scientific and engineering simulations?
 - a) Embedded computer
 - b) **Supercomputer**
 - c) Server

2. Which type of computer is designed to control the engine, monitor the fuel efficiency or manage the entertainment system in a car?
 - a) Smartphone
 - b) Server
 - c) **Embedded computer**

3. Which type of computer is best suited for managing email accounts?
 - a) Database server
 - b) **Email server**
 - c) Game server

4. Which type of computer is designed to be small and energy efficient?
 - a) Desktop
 - b) Supercomputer
 - c) **Embedded computer**

5. Which type of computer is optimized for high performance, reliability and security?
 - a) Supercomputer
 - b) **Server**
 - c) Embedded computer

Activity 2.4

Match each computer with the statement that best describes it.

	Smartphone	Desktop	Laptop
	Supercomputer	Server	Embedded computer
1.	It is a personal computer that is not portable.		<u>Desktop</u>
2.	It is massive and handles one complex task at a time.		<u>Supercomputer</u>
3.	It is small in size, highly reliable and energy efficient.		<u>Embedded computer</u>
4.	It is the smallest and most portable personal computer.		<u>Smartphone</u>
5.	It is used to provide services for other computers.		<u>Server</u>
6.	A personal computer that is portable but not handheld.		<u>Laptop</u>

Activity 2.5

Match each computer with the task for which it is best optimized.

	Desktop	Server	Smartphone
	Supercomputer	Tablet	Embedded computer
1.	Constructing a map of the visible universe		<u>Supercomputer</u>
2.	Getting real-time directions for a destination		<u>Smartphone</u>
3.	Playing an online video game		<u>Desktop</u>
4.	Controlling the doors of an elevator		<u>Embedded computer</u>
5.	Hosting websites for various clients		<u>Server</u>
6.	Drawing an illustration using a stylus		<u>Tablet</u>

Activity 2.6

Answer the questions below about personal computers.

1. Why are personal computers considered general-purpose?

Personal computers are considered to be general-purpose because they are designed to be flexible and to run a wide range of applications. This allows them to perform a wide variety of tasks, such as word processing, browsing the internet, playing games, watching videos and more.

2. “Personal computers are designed for reasonable capability, high flexibility and easy portability at an affordable price.” Is this statement true? Explain.

This statement is true. Personal computers provide the capabilities needed by the average person. Personal computers are designed to be easy to use, even for those who are not computer experts. They are flexible, with a wide range of applications available that can be used for different purposes. They are priced to be affordable for the average consumer, although the cost can vary depending on the specific features and capabilities of the device.

Activity 2.7

Answer the questions below about supercomputers and servers.

1. How would you distinguish between the supercomputer and the server?

1) Purpose: Supercomputers are designed to handle one complex task at a time, such as weather forecasting. Servers are designed to handle many simpler requests from other computers simultaneously.

2) Size: Supercomputers are typically very large and complex. Servers vary in size from large data centers to single computers the size of a desktop.

2. A security breach can happen with any computer system. Compare the impact of a security breach on a server and a supercomputer.

Security breaches are serious threats and can cause significant harm depending on the information being stolen, damaged or lost. A security breach for a server involves the theft, loss or damage to personal, medical or corporate private information stored on the server. A security breach for a supercomputer involves the theft, loss or damage to information related to national security or scientific research. This may compromise the security of the nation or the accuracy and reliability of the scientific calculations and simulations.

Activity 2.8

Answer the questions below about embedded computers. Note that they can be referred to as 'control modules', 'controllers' or 'micro-controllers', among other names.

1. List five common machines or devices in which embedded computers are found.

-
- 1) Home appliances (e.g. refrigerators, washing machines)
 - 2) Medical devices (e.g. pacemakers, MRI machines)
 - 3) Automotive systems (e.g. engine control units, anti-lock brakes)
 - 4) Aircraft systems (e.g. flight control computers, navigation systems)
 - 5) Industrial equipment (e.g. elevators, assembly line machines)
 - 6) Smart home devices (e.g. thermostats, security systems)
 - 7) Wearable devices (e.g. smartwatches, fitness trackers)
 - 8) Consumer electronics (e.g. gaming consoles, digital TVs)
 - 9) ATMs and point-of-sale systems (e.g. credit card readers, cash registers)
 - 10) Traffic control systems (e.g. traffic lights, toll booths)
-

2. Choose one of the machines or devices you listed in Question 1, and describe the function of its embedded computer(s).

-
- 1) Engine control module (ECM) - controls the operation of the engine
 - 2) Anti-lock brake system (ABS) module - controls the brakes
 - 3) Transmission control module (TCM) - controls transmission and gear shifting
 - 4) Airbag control module (ACM) - deploys the airbag in the event of a collision
 - 5) Infotainment system - controls audio and other entertainment functions
 - 6) Climate control module - regulates the air conditioning system
-
-

Activity 2.9

Answer the questions below about reliability. For computers, reliability refers to the ability of a computer to perform its tasks consistently without failure.

1. Compare and discuss the importance of reliability for the operation of a supercomputer used for scientific research, a server handling financial transactions and an embedded computer in an airplane. What would be the adverse consequences of failure?

Reliability is critical for the operation of any computer system, but its

importance can vary depending on the application and context.

1) For a supercomputer used for scientific research, failure could result in

delays in the research, loss of important information, loss of money spent on

the research and loss of trust in the institution conducting the research.

2) For a server handling financial transactions, failure could result in a loss of

money for customers, an opportunity for hackers to breach the system and a

loss of trust in the financial institution.

3) For an embedded computer in an airplane, failure could result in a plane

crash with loss of life of passengers and crew, loss of money as well as loss of

trust in the airlines.

Activity 2.10

Describe a typical day for yourself and the various interactions with computers you may have. Include in your example supercomputers, servers, personal computers and embedded computers.

In the morning, I wake up to the alarm I set on my smartphone (personal computer). I check if I have any messages from my friends (server). Before I get dressed, I check the weather forecast for the day (supercomputer, server). My dad drives me to school in his car (embedded computer). In class, the teacher uses the classroom desktop (personal computer) to project the day's lesson on the board. In the afternoon, I use the microwave oven (embedded computer) to make popcorn. Mom puts laundry into the washing machine (embedded computer). I use my laptop (personal computer) to sign into my school account and check my homework assignments (server). Before bed, I watch television (embedded computer).

LESSON 3: SYSTEMS & INTERFACES

The concept of a system is fundamental to our understanding of ourselves and the world around us. In this lesson, we will explore computers as technological systems and relate them to humans as biological systems, with the commonality being that they are both capable of information processing. The interaction that occurs between computers and humans underlies the technological developments that have led to the tremendous evolution in our way of life.

SYSTEMS & SIGNALS

A system is a set of interconnected components that work together to achieve a particular task or purpose. These parts or components interact with each other in a coordinated manner, allowing the system to function as a cohesive whole.

A SYSTEM IS A SET OF INTERCONNECTED COMPONENTS THAT WORK TOGETHER AS A WHOLE, SERVING A PARTICULAR PURPOSE.

In order for a system to work as a whole, all its parts must interact with each other and coordinate their functions and activities. In the case of systems that depend on information processing, such as humans and computers, this means that the individual parts of a system must be able to communicate with each other; they must be able to exchange information.

Notes

Humans are one example of a biological system capable of information processing. In fact, any living organism with a central nervous system qualifies to be described as an information processing biological system.

A signal is a message that is carried in physical form; it is essentially the transfer of energy in a fashion that carries some meaningful content interpretable by some information processing system.

A SIGNAL IS A MESSAGE CARRIED IN PHYSICAL FORM.

In computers, the form of energy that typically acts as a signal is electrical energy; more precisely, it is electrical potential energy. Humans also use electrical potential energy for internal signaling. In both cases, changes in voltage are propagated along pathways that connect the various parts of the system. For computers, these signals are called digital electronic signals, and they are propagated along conducting wires. Whereas for humans, these signals are called action potentials, and they are propagated along nerves.



Action potential
signals



Digital electronic
signals

Signals play a crucial role in the coordination of the internal components of a system, as they enable communication between different parts and ensure proper functioning.

**SIGNALS ENSURE COMMUNICATION AND COORDINATION BETWEEN
THE PARTS OF A SYSTEM.**

INTERFACES & TRANSLATION

The boundary of a system where it interacts with the outside world is called its interface. For information processing systems, interaction entails an exchange of signals. Signals going into the system are referred to as input, and in response, the signals that come out are output.

THE INTERFACE IS THE BOUNDARY AT WHICH A SYSTEM INTERACTS WITH THE EXTERNAL WORLD.

Every information processing system has its own form of internal signals; humans use action potentials, and computers use digital electronic signals. These are the only signals they can use to communicate and process internally.

External signals of the outside world are quite different from internal signals; they must be converted into internal signals. Light, sound, pressure and any other signal of the outside world must be converted into either action potentials or digital electronic signals. This conversion from one form of signal to another must preserve the integrity of the message itself. In a sense, it is quite analogous to translation between languages, wherein the information conveyed in one language must be preserved, but the language in which it is conveyed is changed. Signal conversion occurs at the interface.

SIGNAL CONVERSION OCCURS AT THE INTERFACE.

INPUT DEVICES

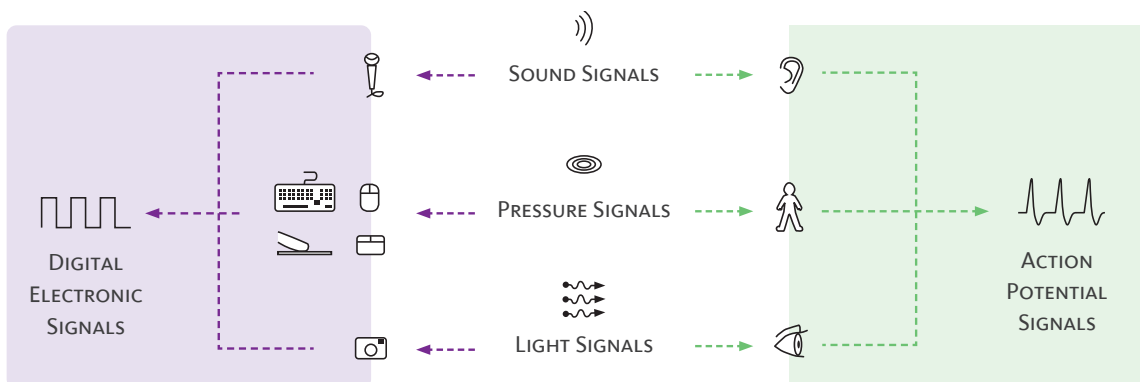
Input devices are the parts of the system that handle information coming in. External signals are converted to internal signals while preserving the infor-

mation they carry. This ensures that the system gets accurate information about its surroundings. Each device is dedicated to converting a specific type of external signal into the required internal signal.

THE INPUT DEVICE CONVERTS A SPECIFIC FORM OF EXTERNAL SIGNAL INTO THE INTERNAL SIGNAL OF THE SYSTEM.

Computer parts responsible for input are called input devices and are often referred to as sensors. Each device is dedicated to the task of converting one form of signal into electronic digital signals. Humans, similarly, have sensory organs which are responsible for accepting input and converting one form of external signal into internal action potential signals.

Sound, pressure and light are commonly converted external signals by both computers and humans. Computers have a microphone and camera to convert sound and light, respectively, while humans have ears and eyes. Computers have various input devices that are capable of converting pressure depending on the type of computer; these are the keyboard, mouse, touchpad and touchscreen. Humans have the skin as a sensory organ capable of converting pressure signals into action potentials.



Sound, pressure and light are commonly converted external signals by both computers and humans. The computer has various input devices, such as the microphone, camera, keyboard, mouse, touchpad and touchscreen, to convert sound, light and pressure into internal signals. In comparison, the human body has the ear, eye and skin, respectively.

In addition, many smartphones are equipped with input devices or sensors that can detect position and motion. Some of the most common sensors used for this purpose include:

1. **Accelerometers** are used to detect linear motion in any of the 3 dimensions: up or down, left or right and forward or backward.
2. **Gyroscopes** are used to detect rotational motion in any one of the 3 dimensions: rotating clockwise or counterclockwise, rolling left or right and pitching up or down.
3. **Magnetometers** are used for magnetic fields. They are used to detect the orientation of the smartphone relative to the Earth's magnetic field.
4. **GPS** are sensors that use satellite signals to determine the location of its computer or device on Earth.

These four sensors work together to determine speed, direction, orientation, position and location, allowing the smartphone to track its own movement and its relative position in Space. This makes them useful in a variety of applications, such as games, fitness trackers, maps and navigation.

In comparison, while humans do not have sensory organs that can detect the Earth's magnetic field or GPS satellite signals, their vestibular system provides highly accurate information about linear movement, rotational movement, orientation and position in space. We can easily tell if we are moving and in which direction. We can also easily tell if we are leaning in any direction relative to the ground.

Furthermore, humans have additional senses. The skin not only provides pressure information, but it also provides information about temperature and pain. In addition, humans are capable of detecting chemical signals through taste and smell.

OUTPUT DEVICES

In contrast to input devices, output devices are responsible for responding to the outside world. Responses, or output, can be in the form of a signal or an action, each of which requires a dedicated output device. In either case, an internal electronic digital signal is converted to an external signal or an external action. This is similar to the case of humans in whom an internal action potential is converted to an external signal, such as the sound of speech, or an external action, such as the movement of a limb.

THE **OUTPUT DEVICE** CONVERTS THE INTERNAL SIGNAL OF THE SYSTEM TO A SPECIFIC FORM OF EXTERNAL SIGNAL OR ACTION.

Computers mainly output two forms of signals: light from the screen display and sound from the speaker. In addition, the handheld computers, namely the tablet and smartphone, can also vibrate, emitting what is called a haptic signal or pressure that can be felt by whoever is holding the computer.

Output devices that respond by performing an action are called actuators; they are capable of moving. Actuators are part of other machines or devices that can be connected to a computer or are controlled by an embedded computer. The printer is a machine that connects to a computer and contains an actuator. The printer's printhead is an actuator that moves across the paper, printing line by line. Other examples of actuators include motors and valves, which are used in various machines such as home appliances, vehicles, manufacturing equipment and robots.

THE **ACTUATOR** IS AN OUTPUT DEVICE CAPABLE OF PERFORMING AN ACTION.



HARDWARE & SOFTWARE

The human computer interface is not just a physical one: it is also a logical one. The hardware component of the interface includes the various physical components, the input and output devices, which are responsible for transmitting and receiving signals. The hardware of the computer handles the physical layer of the interface, which are the signals it is transmitting and receiving.

THE PHYSICAL LAYER IS THE SIGNALS; IT IS HANDLED BY THE HARDWARE.

The software component of the interface determines the information (logic) to be carried by the signals. The software component includes the applications and other programs that determine how the computer behaves and what information it shows the user. The content displayed on the screen is determined by software. What happens when a user clicks a button is determined by software. Everything a computer does is determined by software.

So when we interact with computers, we are interacting with both hardware and software. At the human computer interface, an exchange of signals carrying information is occurring, with the signals being the physical layer and the information being the logical layer.

THE LOGICAL LAYER IS THE INFORMATION IN THE SIGNALS; IT IS HANDLED BY THE SOFTWARE.

EXAMPLE 3.1

When two humans have a conversation, they are interacting. This interaction occurs in both physical and logical layers. In the physical layer, there is an exchange of sound signals, which are sensed by the ears and emitted by the mouth. However, the content of the conversation, the actual information represented by these sound signals, is what forms the logical layer. The logical layer contains the meaning carried by the sound signals.

USER INTERFACE

The user interface (UI) is a term that refers to the software component of the human computer interface, also referred to as the software interface. The UI is responsible for translating user input into instructions that the computer can understand and execute, as well as displaying the results of those instructions for the user.

THE USER INTERFACE IS THE SOFTWARE COMPONENT OF THE HUMAN COMPUTER INTERFACE.

There are several types of user interfaces that evolved over time; these include in chronological order:

1. **Command-Line Interface (CLI):** A CLI requires users to type commands into a text-based interface to perform functions. CLI interfaces are commonly used in programming and system administration but are challenging for users unfamiliar with programming or technical jargon. Examples include Windows Command Prompt and macOS Terminal.

```
Last login: Wed Apr 13 15:35:52 on ttys001
computer-name:~ username$
```

The Terminal window on macOS showing a standard output typical display. The first line indicates the date and time of the last login. The second line displays the name of the computer (in this case, “computer-name”) followed by the username (in this case, “username”) and a dollar sign (\$), indicating that the terminal is ready to accept commands.

2. **Graphical User Interface (GUI):** A GUI uses graphical elements such as windows, icons and menus to represent different functions and commands. The GUI made computer systems more accessible and user-friendly by reducing the need for users to remember complex commands and by providing a more intuitive and visual interface. Examples include Windows and macOS.

Graphical Elements of the GUI

The GUI has three common graphical elements: windows, icons and menus. A window is a separate area of the screen containing content related to a specific task or application. More than one window can be opened, allowing users to interact with multiple applications or tasks simultaneously. In addition, icons and menus allow users to select applications, files and commands.

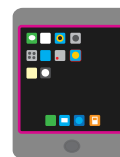
THE GRAPHICAL USER INTERFACE USES WINDOWS, ICONS AND MENUS TO REPRESENT DIFFERENT FUNCTIONS AND COMMANDS.



Desktop



Laptop



Tablet



Smartphone

Desktops and laptops have pointing devices, the mouse and touchpad, respectively. These hardware components are used to control a software component displayed on the screen, the pointer. Tablets and smartphones have touchscreens, a technology that allows pointing to be replaced by touching.

There are two main types of GUIs that are distinguished based on the technology they incorporate to select and control graphical elements: the WIMP and touchscreen interfaces.

WIMP Interface: The acronym WIMP stands for windows, icons, menus and pointer. A pointer is a graphical element, typically in the form of an arrow or hand, used to interact with the interface by selecting icons and items in menus and windows. It is controlled by a mouse or touchpad, both of which are input devices known as pointing devices.

Touchscreen Interface: With the rise of mobile devices, the touchscreen interface became popular, allowing users to interact with devices directly by touching the screen using natural gestures such as swiping, tapping, or pinching, making it easy for users to learn and use. Touchscreens kept the windows, icons and menus but eliminated the need for pointing devices, such as the mouse or keyboard.

3. **Voice User Interface (VUI):** Most recently, the (VUI) has become more prevalent, allowing users to interact with devices using natural language and voice commands. VUIs are becoming increasingly popular in smart home devices and personal computers, such as Apple's Siri, Amazon's Alexa, Google's Assistant and Microsoft's Cortana.

Career Tips

Human Computer Interaction is a multidisciplinary field that combines principles from computer science, cognitive psychology, design and other fields to study how people interact with computers and how to best design user interfaces.

USABILITY

Usability is a design feature that refers to how easy it is for users to interact with a computer in order to complete a particular task. The evolution of the user interface and its related hardware components reflects an evolution in usability.

USABILITY IS THE EASE WITH WHICH A DEVICE CAN BE USED TO COMPLETE A PARTICULAR TASK.

The touchscreen interface is often considered highly intuitive; users can simply touch the screen to interact with it, which many people find easy to learn and use regardless of their knowledge and skills with computers.

The WIMP interface requires a mouse or touchpad, which may require some learning and practice for users who are unfamiliar with the technology. However, once users become familiar with the interface, it can be quite efficient and effective for many computing tasks.

The CLI requires more learning and memorization of specific commands on the part of the user. As a result, it is generally only usable by IT professionals or those with advanced computing skills.

The VUI is an interface that has emerged in recent years. It allows users to interact with computers through speech. While it may seem a step up in usability, this interface is still in its early stages of development.

Overall, the usability of a user interface is an important consideration in designing effective and efficient human-computer interaction. The evolution of hardware components and the user interface itself has reflected a trend towards more intuitive and easy-to-use interfaces, which has allowed a wide range of people to use computers.

SUMMARY

A system is a set of interconnected components that work together as a whole, serving a particular purpose. Humans and computers are examples of information processing systems.

A signal is a message carried in physical form. In an information processing system, signals ensure communication and coordination between the parts of the system. Every system has its own specific type of internal signal, e.g. humans have action potentials, and computers have digital electronic signals.

The interface is the boundary at which a system interacts with other systems and the external world. Signals must be converted between external signals and internal signals. This conversion occurs at the interface and must preserve the integrity of the information carried by the signals.

The interface of a system comprises both input and output devices. Input devices convert external signals into internal signals. Output devices convert internal signals into external signals. Each device is dedicated to a specific type of external signal.

Some output is in the form of action. The output device that performs this action is referred to as an actuator.

Information exists in two layers: the physical and virtual layers. The physical layer contains the actual signals and is handled by the hardware. The virtual layer is the information represented in the signals and is handled by the software.

The user interface is the software component of the human computer interface. This interface has evolved from a command-line interface to a graphical user interface and, more recently, a voice user interface.

The graphical user interface uses windows, icons and menus to represent different functions and commands. The WIMP interface and touchscreen interface are both types of graphical user interfaces. They both have windows, icons and menus, but the WIMP interface includes either a mouse or touchpad, which the touchscreen interface replaces with a touchscreen.

Usability is the ease with which a device can be used to complete a particular task.

A directory is a logical construct used by the filing system to organize files in a computer. A directory is used to hold files and, in many cases, other directories, referred to as subdirectories. The term directory is synonymous with “folder”.

Activity 3.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>T</u>	A system is a set of interconnected components that work together to achieve a particular task or purpose.
<u>F</u>	Humans and computers do not share any common features as systems.
<u>F</u>	Signal conversion at the interface does not need to preserve the integrity of the message.
<u>T</u>	Input devices convert external signals into internal signals while preserving the information they carry.
<u>T</u>	Both humans and computers are capable of converting sound, pressure and light signals.
<u>F</u>	Actuators are only used in computers and not in other machines or devices.
<u>F</u>	The human-computer interface only includes hardware components.
<u>F</u>	Command-Line Interface (CLI) is the most user-friendly type of interface.
<u>T</u>	The Graphical User Interface (GUI) made computers more accessible by providing a more intuitive and visual interface.
<u>F</u>	The Voice User Interface (VUI) is a highly developed and mature interface technology.

Activity 3.2

Answer the questions below.

1. What do computers and humans have in common as systems?
 - a) They both use digital electronic signals.
 - b) **They are both capable of information processing.**
 - c) They both use action potentials.

2. What is the purpose of an interface in an information processing system?
 - a) To store information
 - b) To create new signals
 - c) **To convert external signals into internal signals**

3. Which form of energy typically acts as a signal in computers?
 - a) Thermal energy
 - b) **Electrical potential energy**
 - c) Kinetic energy

4. In the context of information processing systems, what is a signal?
 - a) A physical form of a message
 - b) A method of storing data
 - c) **A type of energy conversion**

5. Signals going into an information processing system are referred to as:
 - a) Output
 - b) **Input**
 - c) Interface

Activity 3.3

Answer the questions below.

1. What is the primary function of input devices?
 - a) Converting internal signals to external signals
 - b) **Converting external signals to internal signals**
 - c) Storing signals

2. Which two devices are responsible for converting sound signals?
 - a) Microphone - Keyboard
 - b) Mouse - Speaker
 - c) **Microphone - Speaker**

3. What type of sensor is used to detect linear motion in smartphones?
 - a) Gyroscope
 - b) **Accelerometer**
 - c) Magnetometer

4. What type of output device is responsible for performing an action?
 - a) Sensor
 - b) **Actuator**
 - c) Transducer

5. Which of the following is an example of an actuator in a printer?
 - a) Ink cartridge
 - b) Paper tray
 - c) **Printhead**

Activity 3.4

Answer the questions below.

1. The human-computer interface consists of:
 - a) Both input and output devices
 - b) Only input devices
 - c) Only output devices

2. Which type of interface uses elements such as windows, icons and menus?
 - a) Command-Line Interface
 - b) Voice User Interface
 - c) Graphical User Interface

3. Which interface allows users to interact with devices using natural gestures?
 - a) WIMP Interface
 - b) Touchscreen Interface
 - c) CLI Interface

4. The term “usability” refers to:
 - a) How easy it is for users to interact with a computer
 - b) The amount of memory a computer has
 - c) The processing power of a computer

5. Which interface is considered the most user-friendly for non-technical users?
 - a) Command-Line Interface (CLI)
 - b) Graphical User Interface (GUI)
 - c) Voice User Interface (VUI)

Activity 3.5

Answer the questions below about signal conversion at the interface.

1. What is the purpose of signal conversion at the interface of an information processing system?

The purpose of signal conversion at the interface is to transform external signals from the outside world into internal signals while preserving the integrity of the message. This conversion allows the system to use and respond to the information it receives from the external environment.

2. Why is preserving the integrity of the message important during signal conversion at the interface?

Preserving the integrity of the message during signal conversion at the interface is crucial because it ensures that the information conveyed by the signal remains accurate and meaningful when it is transformed into the internal signals used by the information processing system. This allows the system to correctly interpret and respond to the input it receives from the external environment.

Activity 3.6

Answer the questions below about humans and computers.

1. State the similarities that exist between humans and computers.

They are both systems.

They both can process information.

They both can sense and respond to their environment.

They are both capable of signal conversion at their interface.

They both use electrical potential energy for internal signaling.

2. “The ability to gather information from the environment adds functionality to a system and enhances its chances of success.”

Comment and give an example.

The more information a system can gather from its environment, the better

its chances of navigating through its environment to find sources of energy

and shelter, as well as to avoid obstacles and threats. Each sense adds a new

set of information. Therefore, adding senses or input devices adds more

information. The sense of smell can help someone avoid a fire before seeing it.

The sensing of Earth’s magnetic field helps birds migrate to warmer climates.

Having a smartphone with more sensors makes it more useful.

Activity 3.7

Answer the questions below about input and output devices.

1. What does the term “actuator” refer to in technology? Provide an example.

An actuator is an output device that converts internal digital electronic signals to movement. An example is the motor that drives the turntable of a microwave oven.

2. Identify and list the input devices on your personal computer.

Windows: 1) Right-click on “Start”, 2) select “Power User Menu”, 3) select “Device Manager” and then search the listed hardware categories.

macOS: 1) Click on the Apple logo, 2) select “About This Mac”, 3) select “System Report” or “More Info” and then search the listed hardware categories.

Androids & iPhones: A third-party app, such as “Sensors Multitool” or “Sensors Toolbox” will provide device-sensor information.

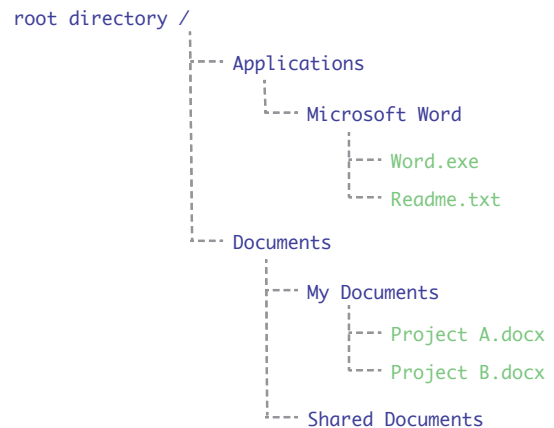
DIRECTORIES

Any information stored in a computer is stored in the form of a file. All files are organized and managed by the computer's filing system. The filing system uses the idea of a folder to organize files. A folder acts as a virtual container used to hold files and, in many cases, other folders, referred to as subfolders.

A more technical term for folder is directory. To restate using this term, the filing system organizes files under directories. A directory is used to hold files and, in many cases, other directories, referred to as subdirectories. This organization of directories within other directories is referred to as a hierarchical structure.

In this example, the directory structure shows the root directory and two levels of subdirectories.

In addition, four example files are shown that appear in green with their file extensions.



In the GUI, the folder icon is a graphical element of the user interface used to represent a directory. It is a visual abstraction of the more complex filing system managed by the computer. It is a simplification that allows a wide range of users to use the computer without having to deal with complex filing systems.

Notes

Abstraction is a fundamental concept in many fields, including computer science, sciences, mathematics and philosophy. It allows us to focus on the essential characteristics of a concept or system without getting bogged down in irrelevant details and is, therefore, a key aspect of problem-solving, innovation and creativity.

Activity 3.8

Use the command line interface to create a file in a specific folder. Then, find and add content to this file through the graphical user interface. Compare the two interfaces.

1. In the CLI, create a text file in the Documents directory and name it “My File” (see instructions). Add content, then save and close the file.
2. In the GUI, open “My File”, add additional content, then save and close the file.
3. Compare the two interfaces and their usability.

Instructions for macOS

1. Open Terminal, and list the directories under the root directory using the ‘ls’ command.

```
ls
```

2. Change directory using the ‘cd’ command followed by the name of the destination directory, which in this case is ‘Documents’.

```
cd 'Documents'
```

3. List the contents of the directory sorted in alphabetical order using the ‘ls | sort’ command.

```
ls | sort
```

4. Create a text file using the ‘touch’ command followed by the name of the file and its extension.

```
touch 'My File.txt'
```

5. Open the text file using the ‘nano’ command followed by the name of the file and its extension.

```
nano 'My File.txt'
```

6. Add text by simply typing the text, then save and close using ‘Ctrl+X’ and then ‘Y’ followed by ‘Enter’.

7. List the contents of the directory again to check that the file was created.

```
ls | sort
```

Instructions for Windows

1. Open Command Prompt, and list the directories under the root directory using the ‘dir’ command.

```
dir
```

2. Change directory using the ‘cd’ command followed by the name of the destination directory, which in this case is ‘Documents’.

```
cd "Documents"
```

3. List the contents of the directory sorted in alphabetical order using the ‘dir’ command with the ‘/O:N’ option.

```
dir /O:N
```

4. Create a text file using the ‘notepad’ command followed by the name of the file and its extension to open a file in the Notepad app.

```
notepad "My File.txt"
```

5. Add text by simply typing the text, then save and close using ‘Ctrl+S’ then ‘Ctrl+W’.

6. List the contents of the directory again to check that the file was created.

```
dir /O:N
```

Activity 3.9

Use the command line interface and the graphical user interface to create a backup folder that contains copies of your original files. Compare the two interfaces.

1. In the CLI, create a new directory under Documents. Name the new directory My Backups and move a copy of the file you created in Activity 3.8 into this directory.
2. In the GUI, create another directory under Documents. Name the new directory My Backups 2 and move a copy of the file you created in Activity 3.8 into this directory.
3. Compare the two interfaces and their usability.

Instructions for macOS

1. Open Terminal, and navigate from the root directory to the 'Documents' directory.

```
cd 'Documents'
```

2. Make a new directory using the 'mkdir' command. Name the directory 'My Backups'.

```
mkdir 'My Backups'
```

3. Copy the file 'My File.txt' using the 'cp' command. Name the copy 'Copy My File.txt'.

```
cp 'My File.txt' 'Copy My File.txt'
```

4. Move the copied file to the new directory 'My Backups'.

```
mv 'Copy My File.txt' 'My Backups'
```

5. Change directory using the 'cd' command to the 'My Backups' directory.

```
cd 'My Backups'
```

6. List the contents of the directory to check that the file is in it.

```
ls
```

Instructions for Windows

1. Open Command Prompt, and navigate from the root directory to the 'Documents' directory. Replace {your_username} with your actual username.

```
cd C:\Users{your_username}\Documents
```

2. Make a new directory using the 'mkdir' command. Name the directory 'My Backups'.

```
mkdir "My Backups"
```

3. Copy the file 'My File.txt' using the 'cp' command. Name the copy 'Copy My File.txt'.

```
copy "My File.txt" "Copy My File.txt"
```

4. Move the copied file to the new directory 'My Backups'.

```
move 'Copy My File.txt' 'My Backups'
```

5. Change directory using the 'cd' command to the 'My Backups' directory.

```
cd 'My Backups'
```

6. List the contents of the directory to check that the file is in it.

```
dir
```

LESSON 4: BEHIND THE INTERFACE

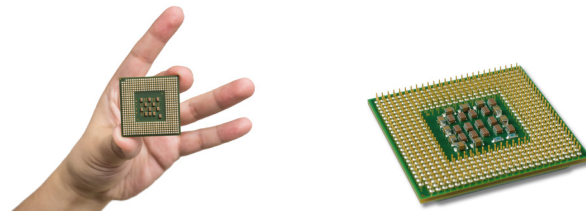
At the interface of an information processing system, only the input and output are visible to the user, whereas the steps that occur inside the system are not. This is because the interface is designed as an abstraction, a simplification of the complex internal workings of the system. Such internal workings are the topic of interest in this lesson.

THE PROCESSOR

In order for a system to generate a response, it must be able to process the information it has gathered. The system must be able to change input to output. This is the task of the processor. The processor is that part of the system dedicated to changing input to output, thereby allowing the system to respond.

THE PROCESSOR IS THE PART OF THE SYSTEM THAT CHANGES INPUT TO OUTPUT.

Just as the user interface is both hardware and software, in the context of processing, we can broadly state that the processor is the hardware through which the signals move and are operated on, while the stored programs are the software that determines the order in which these operations happen and the components of the processor that will perform them.



A typical desktop CPU package measures about 4-5 square centimeters. However, the size of the actual processor chip inside the package is much smaller, typically only a few square millimeters or less.

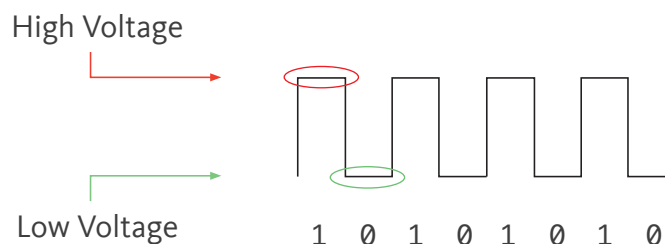
DIGITAL ELECTRONIC SIGNALS

Inside the computer, information is carried as electrical signals. These signals travel through wires arranged in groups of 64 or 32 parallel wires, depending on the computer, with each wire carrying 1 signal or 1 bit of information. A computer with a 64-bit architecture uses 64 parallel wires, while a computer with a 32-bit architecture uses 32 parallel wires, allowing for 64 bits and 32 bits of information to be transmitted simultaneously, respectively.

Each bit is a pulse of electrical potential energy, whose intensity is measured in volts. Therefore, signals can be distinguished based on the intensity of energy they have, into either high voltage or low voltage. To simplify this, a high voltage signal is labeled '1', and a low voltage signal is labeled '0', even though it does not actually have zero volts of energy. Since a signal can only be one of two options, high or low, 1 or 0, we call this type of signal 'binary', which means 'two'. Electrical signals that are binary in nature are referred to as digital electronic signals.

DIGITAL ELECTRONIC SIGNALS ARE BINARY IN NATURE, BEING EITHER HIGH VOLTAGE (1) OR LOW VOLTAGE (0).

A sequence of 8 bits is referred to as a 'byte', whereas the term 'word' is used to refer to the maximum number of bits that can be transmitted simultaneously. In a 32-bit processor, a 'word' is 32 bits long, whereas in a 64-bit processor, a 'word' is 64 bits long.



This is an 8-bit sequence of signals with alternating high and low voltages. The byte can be represented in binary as a series of alternating 1's and 0's: 10101010.

LOGIC CIRCUITS

A logic circuit is a collection of electronic components, mainly transistors, connected by wires in specific configurations. The configuration of transistors determines the operation the logic circuit can perform. Changing the configuration changes the operation. Therefore, a logic circuit that performs addition operations would have a different configuration of transistors from a circuit that performs multiplication operations.

A LOGIC CIRCUIT IS A COLLECTION OF ELECTRONIC COMPONENTS DESIGNED TO PERFORM A PARTICULAR OPERATION.

Processing occurs when digital electronic signals pass through a logic circuit. For example, the adder is a logic circuit that performs an addition operation. It receives two input signals, adds them and then outputs one signal, their sum. There are other logic circuits in the processor, such as the multiplier, comparator and multiplexer.

LOGIC CIRCUIT	PURPOSE
ADDER	Adds 2 inputs
MULTIPLIER	Multiplies 2 inputs
SHIFTER	Shifts the bits of 1 input by 1 or more places
COMPARATOR	Compares 2 inputs as equal, less or greater than
MULTIPLEXER	Selects 1 from several inputs

THE OPERATION A LOGIC CIRCUIT PERFORMS IS DETERMINED BY THE CONFIGURATION OF ITS COMPONENTS.

EXAMPLE 4.1

Let's assume there is a small processor with a 4-bit architecture instead of the more common 64 bits. This hypothetical processor has 4 logic circuits: the adder, multiplier, shifter and comparator.

Let's also assume that the processor receives two 4-bit signals as input, 0110 and 0001. If these two inputs pass through the adder, the output would be their sum, 0111.



Two input signals, 0110 and 0001, pass through an adder logic circuit. The circuit performs the addition operation and generates an output which is the sum, 0111. Note that addition follows the same rules in binary as it does in decimal.

	0	1	1	0
+	0	0	0	1
	0	1	1	1

Now, let's pass the same 4-bit signals, 0110 and 0001, through the multiplier, the output would be their product, 0110. This happens because the logic circuits are arranged in a manner that performs a multiplication operation rather than an addition.



The same two input signals, 0110 and 0001, pass through a multiplier logic circuit. The circuit performs the multiplication operation and generates an output which is the product 0110. Note that multiplication also follows the same rules in binary as it does in decimal. In this case, any number multiplied by 1 is itself.

	0	1	1	0
×	0	0	0	1
	0	1	1	0

On the other hand, the shifter logic circuit would handle the two inputs differently. The first input, 0110, would be considered the signal on which it would perform the shift operation. The second input, 0001, would be the number of places that would be shifted, which in this case is 1 place to the left.



The same two input signals, 0110 and 0001, pass through a shifter logic circuit. The logic circuit performs the shift operation on the first input by shifting its bits by the number of places indicated by the second input, which in this case is 1. The left-most bit is dropped, and a 0 is added as the right-most bit.

In	0	1	1	0
	1	1	0	
Out	1	1	0	0

The comparator accepts two signals but can output 1 of 3 possible outputs: 1) greater than, 2) less than or 3) equal to. It will generate an output signal based on the comparison of the first signal with the second signal. In our small 4-bit processor, we will assign a specific sequence of bits that will be interpreted as greater than, less than or equal to, 1111, 1001 or 0000, respectively.



The same two input signals, 0110 and 0001, pass through a comparator logic circuit. The logic circuit compares them. Since the first signal is greater than the second, $0110 > 0001$, the comparator will output the signal for greater than, 1111.

>	1	1	1	1
=	1	0	0	1
<	0	0	0	0

ALU

The arithmetic logic unit (ALU) is that part of the processor responsible for executing instructions. It contains all the logic circuits needed to perform the operations required for executing instructions.

THE ALU IS THAT PART OF THE PROCESSOR RESPONSIBLE FOR PERFORMING OPERATIONS ON DATA.

An instruction typically specifies the operation that needs to be performed. However, for an operation to function, it requires operands or information to work on. The instruction must provide this information either as data or as an address. Data can be provided immediately in the instruction itself, or the instruction can provide a source address at which the data can be found.

EXAMPLE 4.2

The instruction to add two numbers must specify that the operation to be performed is an addition. It must also specify the two numbers that are to be added, the operands. Let's assume that the numbers to be added are 2 and 3. In this case, the instruction in human-readable language would be:

```
ADD 2,3
```

Notes

Learning the concepts that underpin programming is more critical than learning the programming language itself. Pseudocode is used for that purpose. Written in English, it allows the student to focus on thought processes rather than syntax and semantics.

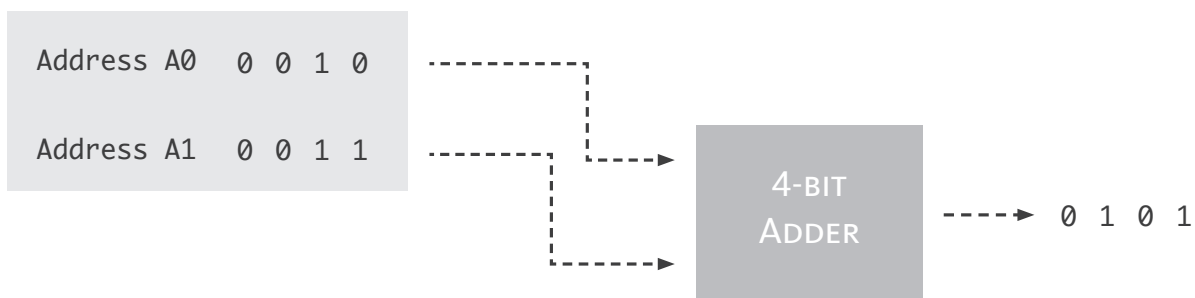


The instruction specifies that the input signals should be 2 and 3, which in binary are represented as 0010 and 0011, respectively. The output from the adder logic circuit would be 5, 0101.

			1	
	0	0	1	0
+	0	0	1	1
	0	1	0	1

Alternatively, the numbers 2 and 3 may already be in the computer and are located at Address A0 and Address A1, respectively. In this case, the instruction would provide the processor with the addresses from where to obtain this data.

ADD A0,A1



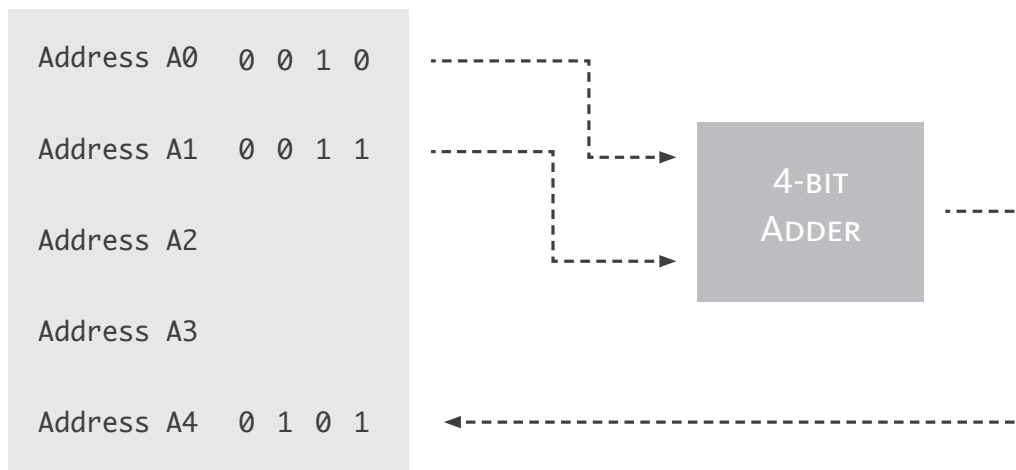
In this case, the instruction provides the data by referring the processor to the source addresses at which the data can be found, Address A0 and Address A1. The data is fetched and fed through the adder logic circuit.

			1	
	0	0	1	0
+	0	0	1	1
	0	1	0	1

However, we still need to figure out where the ALU should send the output of its logic circuits. Just like the ALU has a source for its information, it must

also have a destination. Therefore, the instruction must specify, in addition to a source address, a destination address - the address to which the output should be sent.

ADD A4: A0,A1



The instruction specifies the destination address first, followed by a colon. The two source addresses, Address A0 and Address A1 follow the colon. The processor provides the data from the two source addresses to the adder logic circuit and then delivers the output to the specified address, Address A4.

				1
	0	0	1	0
+	0	0	1	1
	0	1	0	1

Since every instruction must be processed by a logic circuit and the ALU can fit only so many logic circuits, that means that the number of instructions one ALU can execute is limited. In fact, despite the many tasks we perform using our computers, from preparing essays to playing games and watching videos, all these tasks are carried out using a limited set of instructions.

THE INSTRUCTION SET OF A PROCESSOR IS THE SET OF ALL INSTRUCTIONS IT CAN PERFORM.

When designing a processor, engineers first determine the set of instructions the processor needs to execute. Then, they determine the operations needed to execute each instruction and select the logic circuits that perform each operation. Finally, they organize the logic circuits in such a way that allows the operations to be performed in the correct order.

CONTROL

The control unit, CU, is an integral part of the processor. The CU is a collection of logic circuits whose main purpose is to direct the flow of signals through the processor and other parts of the computer. It does so by: 1) decoding instructions to identify the operations that need to be performed, 2) generating control signals to activate the logic circuits that perform these operations and 3) coordinating the operation of the different components of the computer.

In order for the CU to accomplish its purpose, it must be able to identify the operation that needs to be performed; therefore, every instruction in a processor's instruction set must have a unique operation code, opcode. The CU uses this opcode to determine the logic circuits it must activate.

THE CONTROL UNIT DIRECTS THE FLOW OF SIGNALS THROUGH THE PROCESSOR AND COMPUTER.

EXAMPLE 4.3

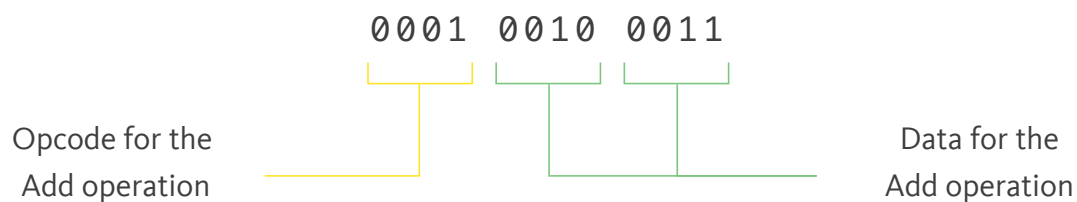
Going back to our 4-bit processor and its four logic circuits, we can create a binary sequence to represent each of the four operation codes required.

Add	0 0 0 1
Multiply	0 0 1 0
Shift	0 0 1 1
Compare	0 1 0 0

Let's revisit the Add instruction that delivers two immediate operands for processing, namely 2 and 3, or 0010 and 0011, respectively. In its human-readable form, it reads:

ADD 2,3

In machine language, using binary signals, this instruction is 12 bits in length and reads:



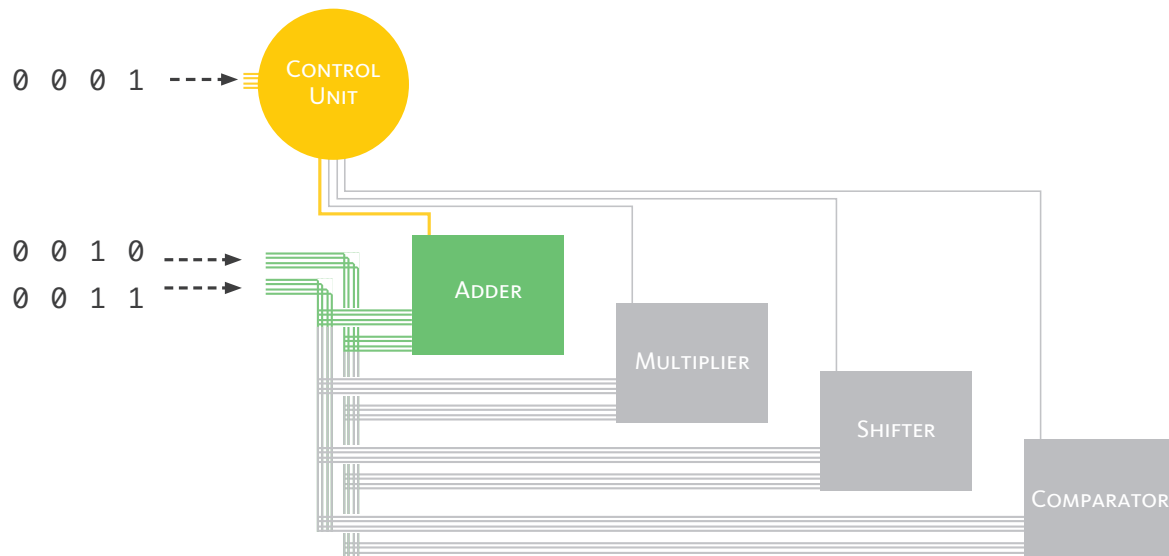
On a side note, this means for our hypothetical small computer to work, 4 bits is not a large enough architecture, and we need to upsize by adding 8 more lines to handle 12 bits instead of 4 bits.

When the instruction is fetched, it is split into its 3 fields: the 4-bit opcode and two 4-bit sequences that represent data. The opcode is sent to the CU for decoding, and, accordingly, a control signal is sent to activate the appropriate logic circuit to which the data is delivered.

MACHINE LANGUAGE IS IN THE FORM OF ELECTRONIC SIGNALS AND IS WRITTEN FOR HUMANS AS ZEROS AND ONES.

Notes

Information exists in two main layers: the logical layer and the physical layer. The logical layer makes sense to us and is represented in the form of human-readable instructions and data. The physical layer is the signals and the logic circuits that process them. The physical layer is the implementation of the logical layer.

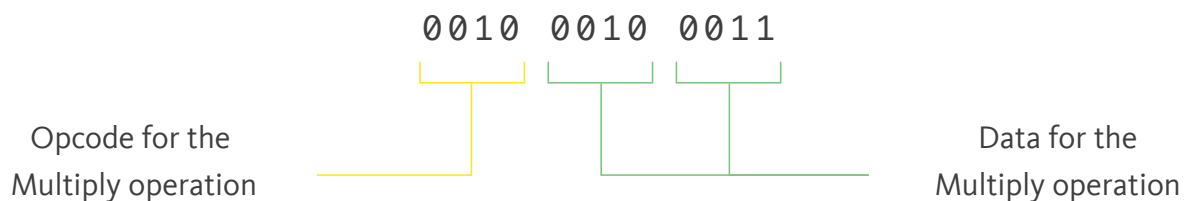


The CU has a control line to each of the four logic circuits it must manage. Once the instruction is fetched, it is split. The opcode `0001` is sent to the CU that interprets it as an Add operation, and then proceeds to send an activation control signal to the adder logic circuit. The activated adder can now receive the two items of data, `0010` and `0011`.

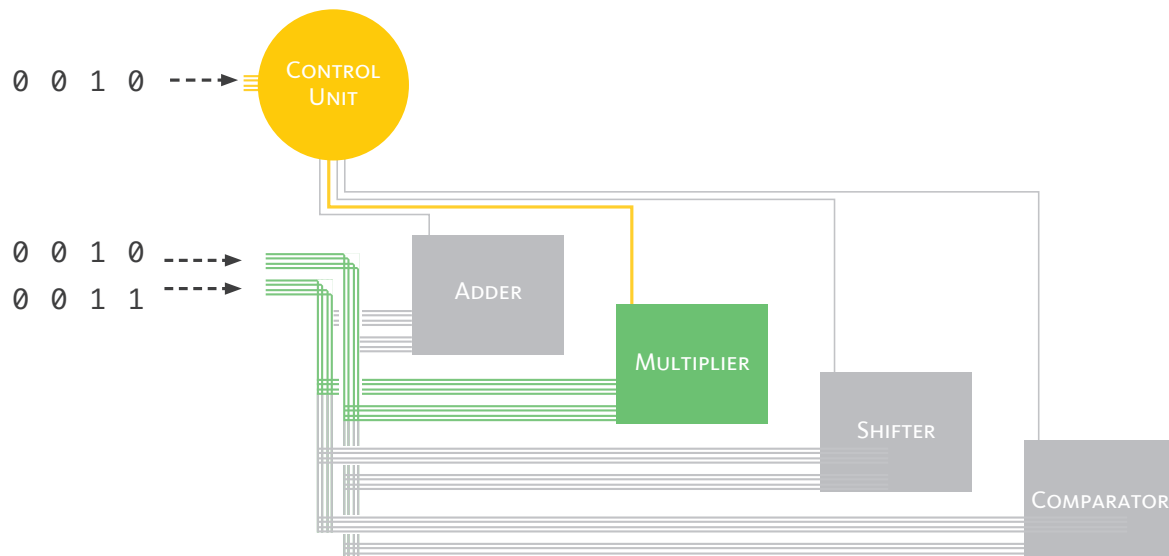
Suppose the Add instruction is changed to a Multiply instruction keeping the same immediate operands, namely 2 and 3, or `0010` and `0011`, respectively. In its human-readable form, the instruction reads:

MULTIPLY 2,3

The opcode for the Multiply operation is `0010`. In machine language, the 12-bit binary instruction reads:



As with every instruction, the instruction is fetched, then it is split into its respective fields. The leftmost field is sent to the CU for decoding, a process that identifies and activates the required components of the computer. The remaining two fields are treated as operands and sent to the activated logic circuit for processing.



In this example, the instruction is for Multiply operation whose opcode is `0010`. The 12-bit binary instruction is split. The 4-bit opcode is sent to the CU, which then activates the multiplier logic circuit by sending a control signal. The activated logic circuit can then receive the two items of data, `0010` and `0011`.

IN MACHINE LANGUAGE, THE INSTRUCTION IS DIVIDED INTO FIELDS, EACH REPRESENTING A PIECE OF INFORMATION.

The previous hypothetical instructions contained immediate operands, which are data immediately delivered to the ALU for processing. However, not all instructions necessarily contain immediate data. In many cases, instructions involve source addresses that indicate the locations where the data is stored. One of the components that hold such data waiting to be processed is the register file.

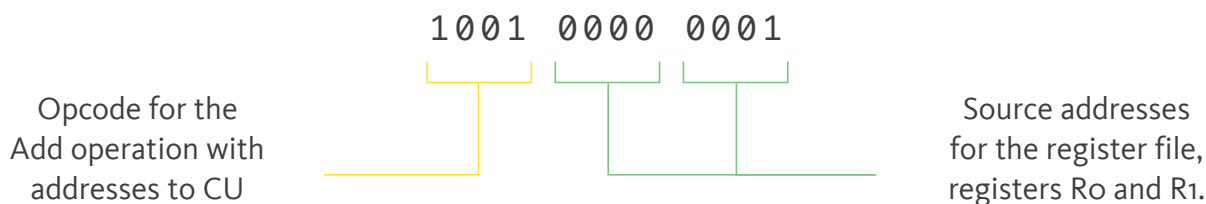
A register is a small and fast memory component within the processor. A processor typically has several registers that are grouped into one working unit referred to as the register file.

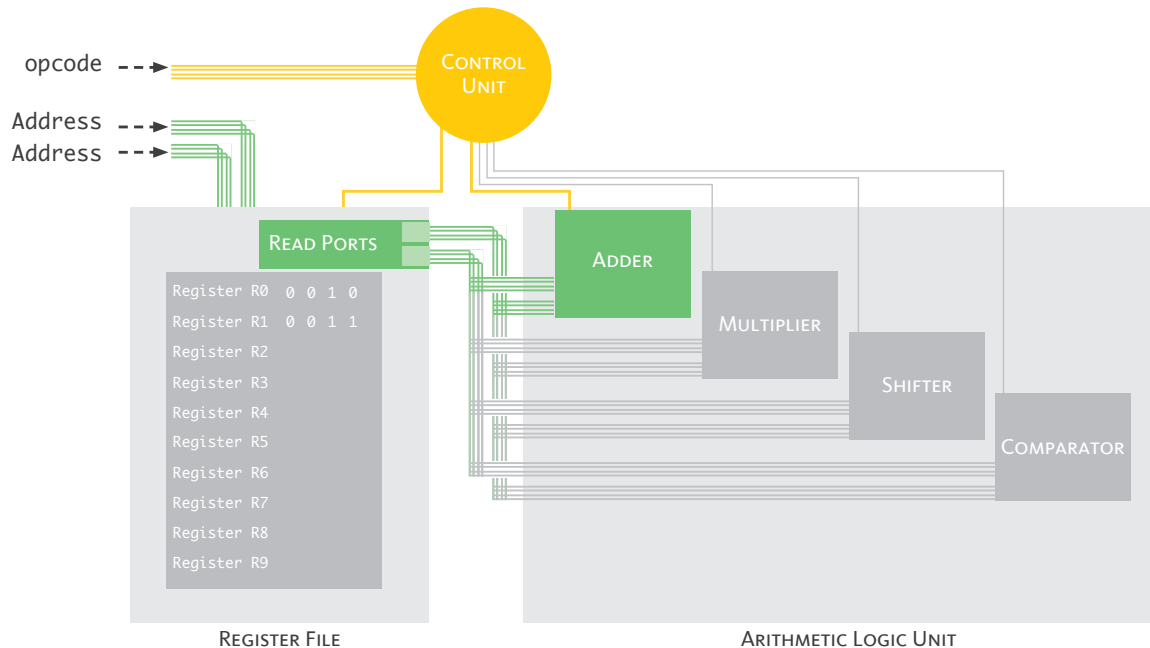
THE REGISTER FILE IS A GROUP OF SMALL AND FAST MEMORY COMPONENTS LOCATED IN CLOSE PROXIMITY TO THE ALU.

Like other components of the processor, the register file is managed by the CU. For instructions that contain source addresses rather than immediate data, the opcode is different. This allows the CU to distinguish between the need to activate the register file and not. When the CU decodes the opcode and identifies that the operation requires the services of the register file, it treats the following two fields as addresses.

A fundamental concept in computing is that any sequence of signals is interpreted based on the context in which it is presented for execution. So if the opcode is for adding immediate data, the following fields are interpreted as numbers. Whereas, if the opcode is for adding the data at source addresses, the following fields are interpreted as addresses.

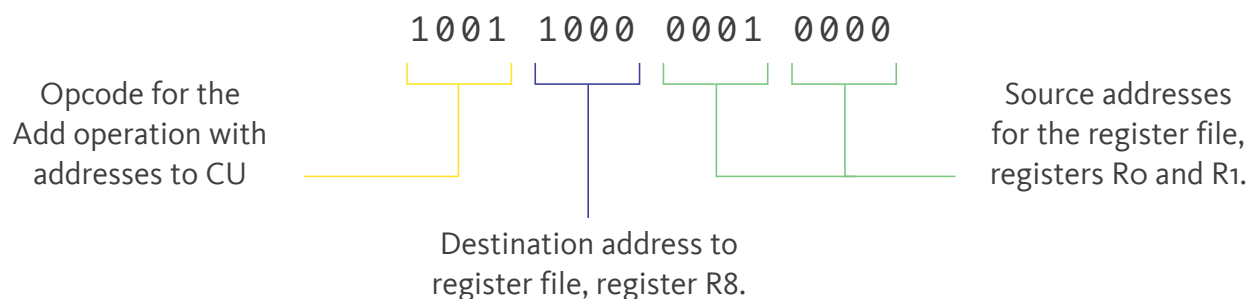
THE INTERPRETATION OF ANY SIGNAL DEPENDS ON THE CONTEXT WITHIN WHICH IT IS BEING EXECUTED.





The instruction is split into 3 fields: the opcode and the two source addresses. The opcode is sent to the control unit, where it is decoded. The CU signals the register file to accept the two addresses as source addresses and to load their contents to the read ports. The adder can then read the data and perform the addition operation.

So far, in our hypothetical processor, we have not considered that data must go somewhere after processing. Once an operation like addition is performed, the result needs to be stored for further use or future instructions. To specify where the output should be stored, the instruction includes a destination address field. This field points to a specific register where the result of the operation will be held.

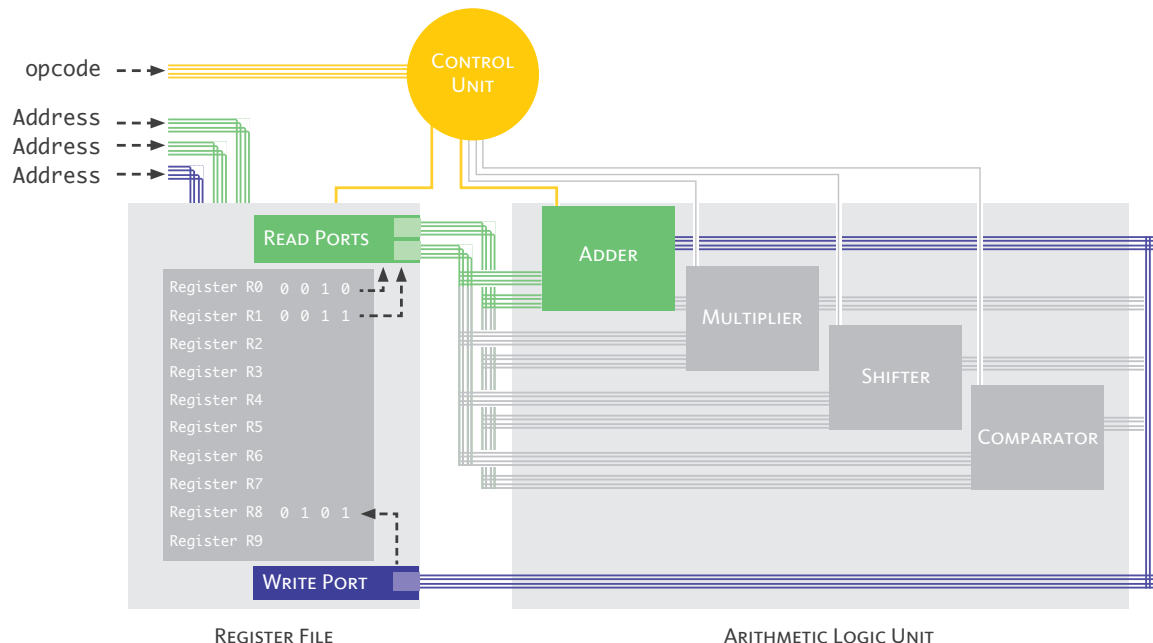


This 16-bit binary sequence, 1001 1000 0001 0000, means: “add the contents of registers R0 and R1 and place the sum in register R8.” In human-readable form, it is transcribed as:

```
ADD R8: R0,R1
```

The destination address, like the source address, typically represents a register number that identifies a specific register within the register file. After decoding the instruction, the control unit signals the register file to write the result into the register specified by the destination address.

By having a designated destination address, the processor ensures that the result is properly stored and can be accessed by subsequent instructions or utilized in further computations.

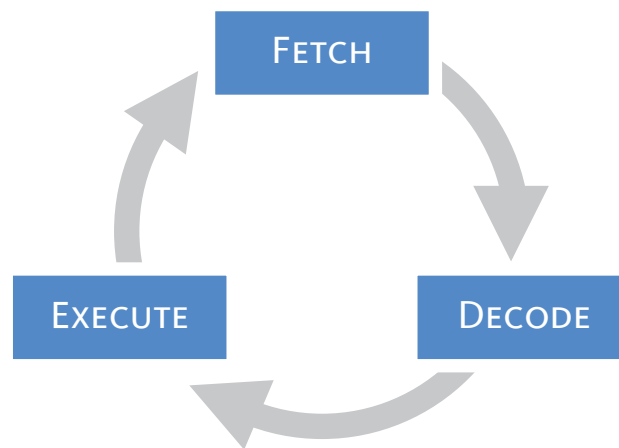


The instruction is split into 4 fields: an opcode, two source addresses and a destination address. The CU signals both the register file and the Adder in the ALU. The register file places the contents of the registers indicated by the source addresses in the read ports. The Adder reads the data, performs the operation and sends the sum back to the register file, which is placed in the register indicated by the destination address.

It is important to note that this lesson reveals the interface between the logical layer, represented in human language, and the physical layer of machine language, represented in digital electronic signals and implemented by logic circuits.

THE INSTRUCTION CYCLE

As long as a processor is working, it is processing instructions one after the other. It fetches an instruction, decodes it, executes it and then fetches the next instruction to repeat the process over and over again. This is referred to as the instruction cycle. Each step is referred to as a stage. Therefore, the instruction cycle is composed of three basic stages: fetch, decode and execute.



THE INSTRUCTION CYCLE IS A SERIES OF REPEATED STAGES: FETCH, DECODE AND EXECUTE.

SUMMARY

While this lesson presented a hypothetical processor, the concepts presented apply to real-life processors.

A computer processes signals referred to as digital electronic signals. These signals are binary in nature in that they can only be one of two values, either high voltage (1) or low voltage (0). Digital electronic signals constitute what is referred to as machine language.

The processor is that part of the computer system that changes input into output. The processor contains logic circuits, which are responsible for processing. Each logic circuit is a collection of electronic components that have a particular arrangement designed to perform a particular operation. The operation performed by a logic circuit is determined by the configuration of its components.

Examples of logic circuits are: adder, multiplier, shifter, comparator and multiplexer.

The processor, also known as the central processing unit (CPU), has three main parts: arithmetic logic unit (ALU), control unit (CU) and register file.

The ALU is responsible for performing operations that include: addition, subtraction, multiplication, division, shifting and comparisons.

The CU coordinates and controls the activities of the processor. It fetches instructions, decodes them and generates the necessary control signals, which ensures that instructions are executed in the correct sequence.

The register file is a group of small and fast memory components called registers. It is located in close proximity to the ALU and is used to hold data and addresses needed in the execution of instructions.

The instruction is split into fields that represent the operation code (opcode), data and/or addresses at which data can be found. The leftmost field is the opcode. The interpretation of the remaining fields depends on the opcode.

The instruction set of a computer is the set of all instructions its processor is designed to perform.

The instruction cycle is a series of stages: fetch, decode and execute. The cycle repeats itself with every instruction.

Activity 4.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>F</u>	The processor is not involved in changing input to output.
<u>T</u>	The main units of a processor are the arithmetic logic circuits, control unit and register file.
<u>T</u>	A digital electronic signal is binary in nature.
<u>T</u>	In a 64-bit processor, a 'word' refers to a sequence of 64 bits.
<u>T</u>	The transistor is an important component of a logic circuit.
<u>F</u>	The configuration of components in a logic circuit does not affect the operation it performs.
<u>T</u>	The ALU is responsible for performing arithmetic operations.
<u>F</u>	The CU is responsible for sending control signals to input and output devices.
<u>T</u>	The opcode determines how the remaining signals in the instruction will be interpreted.
<u>T</u>	Fetch, decode and execute are the three stages of the instruction cycle.

Activity 4.2

Complete the sentences below.

1. Changing input to output is the responsibility of the _____.
 - a) memory
 - b) **processor**
 - c) network adapter

2. Performing arithmetic operations is the responsibility of the _____.
 - a) **ALU**
 - b) CU
 - c) Register file

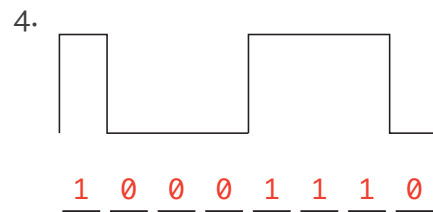
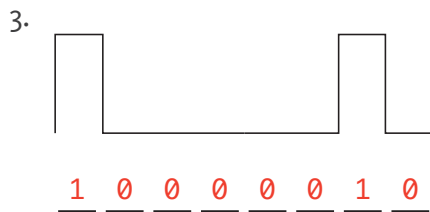
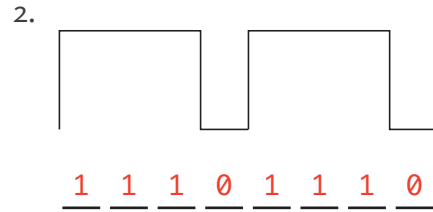
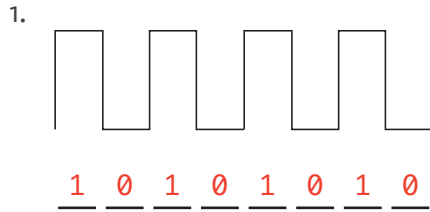
3. Directing the flow of data is the responsibility of the _____.
 - a) ALU
 - b) **CU**
 - c) Register file

4. Which of the following is not true about the register file?
 - a) The register file is a short-term memory unit.
 - b) The register file is a very fast memory unit.
 - c) **The register file is located away from the processor.**

5. Signals going into an information processing system are referred to as:
 - a) decode - fetch - execute
 - b) **fetch - decode - execute**
 - c) execute - decode - fetch

Activity 4.3

Study the digital electronic signals. Encode each in binary form using ones (1) and zeros (0).



Activity 4.4

Perform an addition operation on the 4 pairs of binary numbers.

Note that binary addition is similar to decimal addition, except that $1+1=10$, which requires the 1 to carry forward.

1.

$$\begin{array}{r} 0010 \\ + 0001 \\ \hline 0011 \end{array}$$

2.

$$\begin{array}{r} 0101 \\ + 1000 \\ \hline 1101 \end{array}$$

3.

$$\begin{array}{r} 1100 \\ + 0010 \\ \hline 1110 \end{array}$$

4.

$$\begin{array}{r} 0001 \\ + 0001 \\ \hline 0010 \end{array}$$

5.

$$\begin{array}{r} 11 \\ 0101 \\ + 0101 \\ \hline 1010 \end{array}$$

6.

$$\begin{array}{r} 11 \\ 0011 \\ + 0001 \\ \hline 0100 \end{array}$$

Activity 4.5

Answer the questions below.

1. Which instruction adds two immediates and places the sum in register R8?
 - a) ADD R8: R1,R3
 - b) MULTIPLY R8: 4,5
 - c) **ADD R8: 4,5**

2. Which instruction multiplies data fetched from source addresses and places the product at a destination address?
 - a) **MULTIPLY R5: R4,R0**
 - b) MULTIPLY R8: 4,5
 - c) ADD R8: R1,R3

3. Which instruction is missing a destination address?
 - a) ADD R8: 4,5
 - b) **ADD 2,3**
 - c) ADD R3: R1,R0

4. Which instruction is missing a source address?
 - a) MULTIPLY R6: 0,7
 - b) **MULTIPLY R3: R1**
 - c) MULTIPLY R4: R1,R3

5. Which instruction is missing both source and destination addresses?
 - a) ADD R8: R1,R3
 - b) MULTIPLY R8: 4,5
 - c) **ADD : 4,5**

Activity 4.5

Use the information provided in the tables to translate between human language and the machine of a hypothetical processor to answer the questions.

Opcodes

Add (immediate)	0001
Add (address)	1001
Multiply (immediate)	0010
Multiply (address)	1010

Numbers

0 = 0000	4 = 0100	8 = 1000	12 = 1100
1 = 0001	5 = 0101	9 = 1001	13 = 1101
2 = 0010	6 = 0110	10 = 1010	14 = 1110
3 = 0011	7 = 0111	11 = 1011	15 = 1111

Register Numbers

Register R0 (0000)
Register R1 (0001)
Register R2 (0010)
Register R3 (0011)
Register R4 (0100)
Register R5 (0101)
Register R6 (0110)
Register R7 (0111)

- Which 4-bit binary sequence is designated as the opcode for the operation of adding immediate data?
 - 1001
 - 0001
 - 1101
- Which 4-bit binary sequence represents register R4?
 - 0001
 - 0111
 - 0100
- Which 4-bit binary sequence represents the number 15?
 - 1001
 - 0101
 - 1111

4. Identify the human-readable form of this instruction 1001 0111 0000 0010.
 - a) ADD R4: R1,R2
 - b) MULTIPLY R7: 0,2
 - c) ADD R7: R0,R2

5. Identify the human-readable form of this instruction 0010 0010 0110 0001.
 - a) MULTIPLY R2: 6,1
 - b) MULTIPLY R2: R6,R1
 - c) ADD R2: 6,1

6. Identify the machine language form of this instruction ADD R4: R1,R2.
 - a) 1001 0001 0010 0100
 - b) 1001 0100 0001 0010
 - c) 0001 0100 0001 0010

7. Identify the machine language form of this instruction MULTIPLY R7: 6,7.
 - a) 0010 0110 0111 0111
 - b) 1010 0111 0110 0111
 - c) 0010 0111 0110 0111

8. Write the instruction 1010 0110 0100 0101 in human-readable form.

MULTIPLY R6: R4,R5

9. Write the instruction ADD R3: 4,5 in machine language.

0001 0011 0100 0101

Activity 4.6

Use the information provided in the translation tables of Activity 4.5 to answer the questions.

1. In how many ways can the 4-bit binary sequence 1010 be interpreted? Explain.

It can be interpreted in 3 ways: as an opcode, number and register.

If it is the first field in an instruction, it is interpreted as the opcode for multiply using addresses. If it is the second field, it is interpreted as the register number for the destination register. If it is the third or fourth field, its interpretation will depend on the opcode. An opcode for immediate operations will result in interpreting it as a number, while an opcode for address-related operations will result in interpreting it as a register number.

2. What is the maximum number of integers that can be encoded using 4-bit binary sequences? Can we add more opcodes and registers to our hypothetical processor?

The maximum number of integers that can be encoded is 16, which are the integers 0 to 15. The table in Activity 4.5 lists all the possible 4-bit binary sequences. Alternatively, we can calculate this number. Since there are 4 places and each place can hold one of two values, 0 or 1, we can use $2^4 = 16$.

Yes, we can add opcodes and registers each to a maximum of 16.

Activity 4.7

Answer the questions below.

1. “Instructions provide a context in which signals are interpreted.” Is this statement correct? Explain.

Yes, instructions do provide a context in which signals are interpreted.

Since every instruction has an opcode and the opcode determines how the data is interpreted, then changing the instructions changes the way in which data is interpreted.

2. “I tried to open my Word document using the Excel application, but it didn’t open.” Explain why this statement is true. Does this apply to other applications and files?

In essence, Word and Excel have different ‘instructions’ or ‘opcodes’ for interpreting and handling data, which is why you cannot open a Word document in Excel and expect it to be displayed correctly.

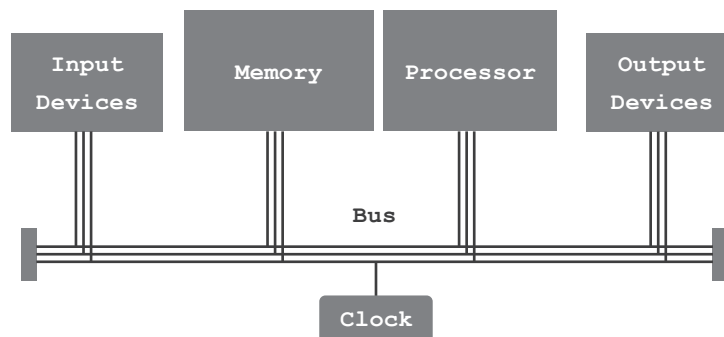
Yes, this principle generally applies to all software applications and their respective files. Each application is specifically designed to work with certain file formats and structures.

LESSON 5: DESIGN OPTIMIZATIONS

Many considerations are taken when designing a system, all in an effort to design the computer to be fast, dependable, energy efficient, easy to use and affordable. Optimizations target design features, such as performance and dependability. This lesson will focus on the strategies used to optimize performance.

ARCHITECTURE

All computer systems contain four basic components: input devices, output devices, a processor and a memory. Input devices and output devices allow the exchange of signals with the outside world. The processor handles the conversion of input to output in accordance with stored programs. The memory is tasked with holding information. All information, whether data or instructions, are held in memory.



The basic architecture of a computer includes a processor, memory components, input devices and output devices. In addition, the bus acts as a communication network connecting these components and a clock to ensure all activities are synchronized.

THE COMPONENTS OF ANY COMPUTER SYSTEM: A PROCESSOR, A MEMORY, INPUT DEVICES, OUTPUT DEVICES, A BUS AND A CLOCK.

THE BUS SYSTEM

The bus, in computing, is a shared communication pathway that allows the components of a computer system to transmit signals to each other. These signals are divided into 3 groups, each carrying one type of information: data, addresses and control signals. Each of these is transmitted on a dedicated set of conducting lines.

Data bus	The set of lines that transmit the actual data being transferred between components. It is bidirectional, meaning it can transmit data in both directions.
Address bus	The set of lines that transmit the addresses used to specify the source or destination of data.
Control bus	The set of lines that transmit bus control signals that are used to coordinate and control the operations of the components within the computer system.

THE BUS IS A SHARED COMMUNICATION PATHWAY THAT CONNECTS ALL THE COMPONENTS OF A COMPUTER SYSTEM.

In a computer system, each component is connected to the bus and assigned an address, a number that uniquely identifies it. The address allows the components to be identified and accessed by the CPU or other components in the system.

When a component, such as the CPU, wants to communicate with another specific component, such as memory or an I/O device, it places the target component's address on the address bus. This address is then transmitted over the bus to indicate the source or destination of the data transfer.

EXAMPLE 5.1

Let's say the CPU is busy processing data for streaming a video that is being viewed on the web browser. With the mouse, the user moves the pointer and selects the window for a Word document essay. The screen switches to the essay window, and the video stops playing. What happened inside the computer?

- | | |
|-----------------|--|
| 1. User | The user moves the mouse and clicks the left button. This generates signals inside the mouse, which are data signals. |
| 2. Mouse | The mouse prepares the data for transmission and places it on the data bus. It places the CPU address on the address bus. It sends an interrupt signal to the control bus. |
| 3. Bus | The bus, based on the address and control signals, delivers the data to the CPU. |
| 4. CPU | The CPU interrupts the process it is working on, streaming the video, in order to process the incoming data. Based on the new data, the CPU switches the active screen from the web browser to the word processor. |

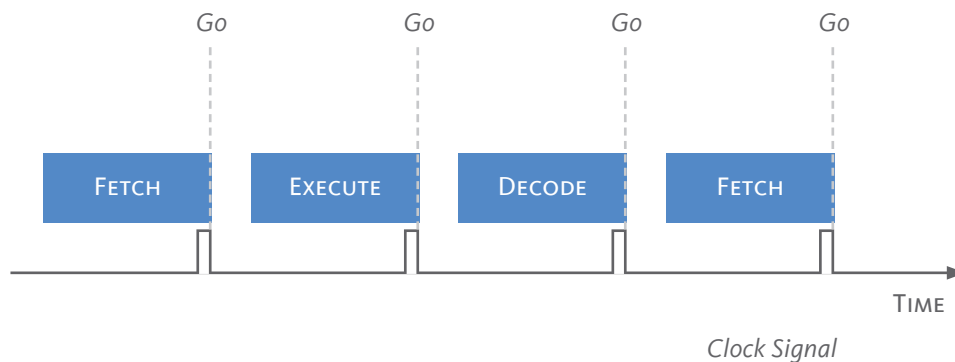
SYNCHRONIZATION

The computer system is a busy place with signals traveling at light speed between the various components while the processor executes one instruction cycle after the other.

Activities occur at different rates. Some components are slower than others, while some instructions take less time to execute than others. This would result in both delays and in errors if left unchecked. Fast operations would pile up, waiting their turn while a slow instruction is being executed, which causes delays and slows performance. Not to mention the errors that can occur if two components try to send data at the same time.

In a synchronized system, all the components operate according to a common time reference, which is typically provided by a time signal generated by an internal clock and distributed to all the components.

The clock has a quartz crystal that oscillates back and forth, vibrates very quickly. Each oscillation is referred to as a clock cycle. With each clock cycle, it generates a timing signal that is placed on the control bus and sent out to all components.

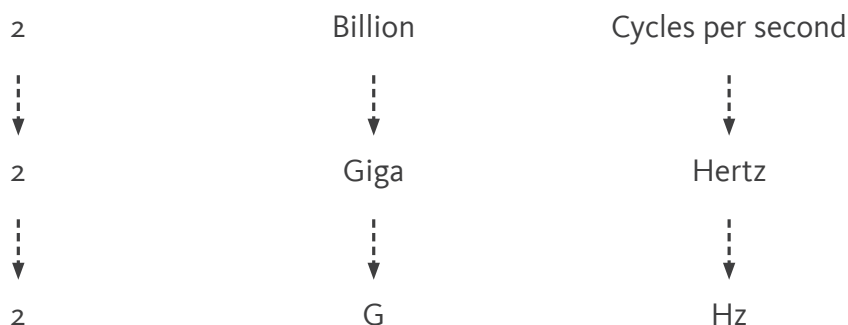


This timing signal is like a “Go” signal; it tells all the components to move their signals to the next stage. This way, all signals move at the same time and don’t “bump” into each other.

Inside the CPU, the clock signals the move from one stage in the instruction cycle to the next. Outside the CPU, the clock signals the transmission of data and addresses on the bus between components. The clock signal is one of the signals transmitted on the control bus.

THE **CLOCK** GENERATES A TIMING SIGNAL THAT SYNCHRONIZES THE TRANSFER OF ALL SIGNALS THROUGHOUT THE COMPUTER.

The processor operates very quickly, mostly due to the high clock frequency. The clock frequency is the number of clock cycles or oscillations that occur per second. For example, if a clock generates 2 billion clock cycles every second, that means its clock frequency is 2 billion cycles per second, or 2 GHz.



Since the clock determines the rate at which the computer operates, then adjusting the clock frequency affects the overall performance of the computer. The clock frequency can be used as a measure of performance, with higher clock frequencies indicating faster performance.

THE CLOCK FREQUENCY IS ONE MEASURE OF COMPUTER PERFORMANCE.

EXAMPLE 5.2

Let's assume there are two hypothetical CPUs, CPU_A and CPU_B. We will make two assumptions for our CPUs, each CPU has 1 processor, and each processor executes only 1 stage of the instruction every clock cycle. If the clock frequency of CPU_A is 6 GHz and that of CPU_B is 3 GHz, then we can say that CPU_A is faster than CPU_B.

$$\frac{6 \times 10^9 \text{ clock cycles}}{1 \text{ second}} \times \frac{1 \text{ instruction}}{3 \text{ clock cycles}} = 2 \times 10^9 \text{ instructions / second}$$

We can also calculate the instruction count per second and compare the two CPUs. If both CPUs are executing 1 stage of the instruction cycle per clock cycle and there are 3 stages, then it will take 3 clock cycles to complete 1 instruction. So CPU_A that runs at 6 GHz, or 6 billion clock cycles per second, will be capable of executing 2 billion instructions per second.

On the other hand, CPU_B is running at 3 GHz or 3 billion clock cycles per second. This is half the speed and is reflected in the calculation, which reveals an instruction count of 1 billion instructions per second.

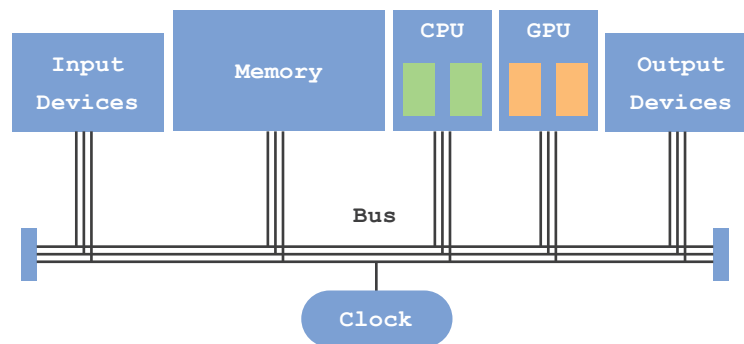
$$\frac{3 \times 10^9 \text{ clock cycles}}{1 \text{ second}} \times \frac{1 \text{ instruction}}{3 \text{ clock cycles}} = 1 \times 10^9 \text{ instructions / second}$$

PARALLELISM

Another strategy for optimizing performance is referred to as parallelism, which means performing tasks simultaneously at the same time. One way this strategy manifests in the design of the computer is in multi-core CPUs, which are CPUs that have more than 1 processor, each referred to as a core.

Another way this strategy is implemented is in the introduction of the graphic processing unit, GPU. The GPU is a processing unit that focuses only on executing instructions that pertain to graphics. It does not concern itself with running the whole of the computer and its apps. In modern-day computers, particularly those used for games, it is common to have both a CPU and a GPU, and each one of these processing units has multiple cores. This is referred to as parallel architecture.

PARALLELISM REFERS TO THE ABILITY TO PERFORM TASKS SIMULTANEOUSLY.



A parallel architecture that includes a GPU operating in parallel with a CPU. The CPU attends to the operation of the whole computer, while the GPU focuses on processing graphics-related instructions.

It is important to note that while a dual-core CPU has two processors, it is faster than a single-core CPU. However, it does not necessarily mean that it is twice as fast. The overall performance of a computer is not limited to clock frequency; it also depends on the task required.

EXAMPLE 5.3

Let's assume CPU_A has a single-core, while CPU_B has a dual-core, and both CPUs are operating at the same clock frequency of 2 GHz. Let's also assume that there is a Program A and that the time it takes for CPU_A to execute Program A is 10 seconds.

If CPU B is tasked with executing Program A, its operating system will have to decide whether it can divide Program A into two parts. If it can, then CPU B can execute both parts simultaneously, each on a core. This would shorten the time to execute Program A from 10 seconds to 5 seconds (assuming both parts are equal).

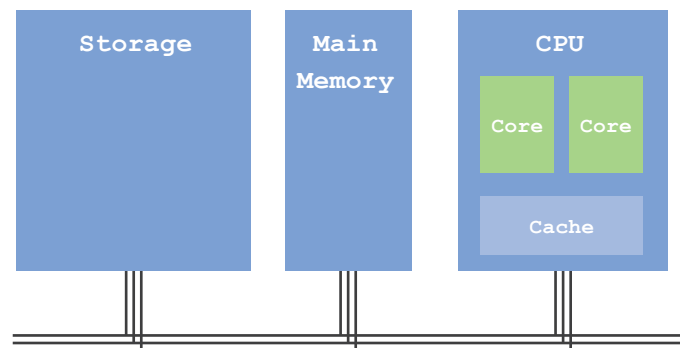
If the operating system cannot divide Program A into two parts, then only one core out of the two will be used, and the execution time remains at 10 seconds.

PROXIMITY

Proximity is another strategy used to optimize computer design. The closer components are together, the better the communication and the faster the transfer of information between them one such implementation of this strategy is the on-chip cache.

Understanding this concept requires an understanding of the memory system and its relation to the processor. Memory is composed of several components that work together as one system, the memory system. The components of this system, in order of size, are: storage, main memory, cache and register file.

THE COMPONENTS OF THE MEMORY SYSTEM ARE: STORAGE, MAIN MEMORY, CACHE AND REGISTER FILE.



The components of the memory system are the storage, main memory, cache and register file. The register file is not illustrated as it is considered part of the core.

The processor depends on memory to supply it with the information it needs. The execution of tasks requires both instructions and data to be delivered promptly from memory to the processor. The overall performance of the computer depends on memory matching its performance with that of the processor. Memory must keep up with the processor; otherwise, the computer's performance will slow down.

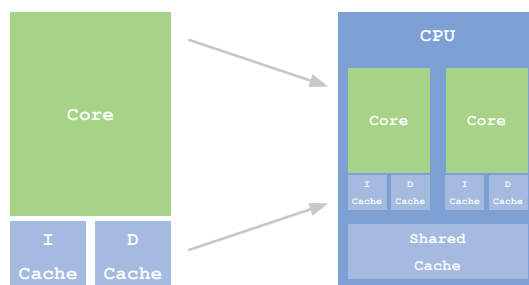
Since close means fast, keeping components, such as the register file and cache, on the same chip as the processor is one way to optimize the performance of the computer. The processor does not have to wait for information to be transmitted on the bus from storage or from main memory. The information it needs is in the registers, and if not there, the information is in the cache. In either case, the information is close by, and the processor does not have to wait.

PROXIMITY OPTIMIZES PERFORMANCE BY PLACING COMPONENTS CLOSE TO THE PROCESSOR.

PREDICTION

Of the optimization strategies, prediction is by far the most ingenious. The ability to predict what comes next greatly improves performance. One such implementation is the separate instruction cache and data cache design.

In this design, the cache is divided into two caches: an instruction cache and a data cache. Having a mixed cache requires the processor to sift through large amounts of information in search of the next instruction. However, we know that instructions follow each other, and therefore, we can predict that if instruction 1 is executed, it will be followed by instruction 2. So, why not arrange all the instructions in order and separate them from the clutter of data?



Each core has two separate caches: one for instructions and the other for data, labeled I-Cache and D-Cache, respectively. The CPU also has a shared cache for all cores on the chip.

Predicting the next instruction allows us to arrange instructions in order and separate them from data. This makes finding the next instruction easy and fast.

PREDICTION OPTIMIZES PERFORMANCE BY PLACING THE MOST LIKELY INFORMATION TO BE USED CLOSEST TO THE PROCESSOR.

Another implementation of prediction is prefetching. When the processor searches for data, it starts by checking the closest memory component to it: the data cache. If it doesn't find the desired data there, it proceeds to check the next component, the shared cache, the main memory, and finally, storage. What's interesting is that when the processor does find the desired data, it doesn't stop at fetching just that particular data item. Instead, it also fetches the adjacent data items. The rationale behind this approach is that if one data item is needed, there is a high chance that the next adjacent data item will be needed soon as well.

By prefetching data that is predicted to be used soon, the processor increases its chances of having the required information available nearby. This proactive data fetching significantly improves the performance of the computer system.

Notes

The principle of locality is fundamental to the operation of the computer. While not explicitly stated as a principle, the lesson clearly identifies both temporal and spatial locality as concepts.

SUMMARY

The components of any computer system are: a processor, a memory, input devices, output devices, a bus and a clock.

The bus is a shared communication pathway that connects all the components of a computer system. The bus is divided into the address bus that transmits source and destination addresses, the control bus that transmits control signals, including the clock timing signal, and the data bus that transmits the actual data.

The clock generates a timing signal that synchronizes the transfer of all signals throughout the computer. Synchronization is a strategy to optimize performance by ensuring that all signals travel without congestion or delay.

The clock frequency is one measure of computer performance. Its unit of measurement is the Hertz. Other measures of performance are the instruction count per second and the program execution time.

Parallel processing refers to the ability to perform tasks simultaneously. Parallelism is a strategy that is implemented in the form of multi-core processors and in the use of graphics processing units that work in parallel with central processing units.

The components of the memory system are: storage, main memory, cache and register file.

Proximity optimizes performance by placing components close to the processor. The closer components are, the faster the exchange of signals. The inclusion of the cache on the same chip as the processor is one implementation of the proximity concept.

Prediction optimizes performance by placing the most likely information to be used closest to the processor. The ability to predict what comes next significantly improves performance. The separation of the instruction and data caches is one implementation of the strategy of prediction. The other is referred to as prefetching, where the processor fetches not only the requested data item but also the adjacent data items. If one data item is needed, there is a high chance that the next adjacent data item will also be needed soon.

Activity 5.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>T</u>	The bus is a component of the computer system.
<u>T</u>	The clock sends out a timing signal to synchronize all components in the computer system.
<u>F</u>	Synchronization is not essential for the proper functioning of the computer..
<u>F</u>	The data bus and the address bus are the only two bus lines.
<u>F</u>	The execution time is not a measure of performance.
<u>T</u>	Performing tasks simultaneously is referred to as parallelism.
<u>T</u>	The CPU and GPU work in parallel.
<u>T</u>	Placing the cache on the same chip as the processor is one implementation of the proximity principle.
<u>T</u>	Prediction is a fundamental concept in computing.
<u>F</u>	Prediction does not improve performance.

Activity 5.2

Complete the sentences below.

1. The _____ is a system of lines that connects all computer components.
 - a) **bus**
 - b) processor
 - c) cache

2. The clock cycle is the time duration between two _____.
 - a) frequencies
 - b) instructions
 - c) **timing signals**

3. The _____ is a measure of computer performance.
 - a) instruction count
 - b) **execution time**
 - c) mean time to failure

4. The unit of clock frequency is the _____.
 - a) Second
 - b) Giga
 - c) **Hertz**

5. The unit of clock cycle is the _____.
 - a) **Second**
 - b) Meter
 - c) Hertz

Activity 5.3

Answer the questions below.

1. The clock is an implementation of which concept?
 - a) **synchronization**
 - b) parallelism
 - c) prediction

2. The cache is an implementation of which concept?
 - a) parallelism
 - b) **proximity**
 - c) synchronization

3. The separate I-cache and D-cache is an implementation of which concept?
 - a) parallelism
 - b) **prediction**
 - c) synchronization

4. The multi-core processor design is an implementation of which concept?
 - a) **parallelism**
 - b) proximity
 - c) prediction

5. The strategy of prefetching is an implementation of which concept?
 - a) **prediction**
 - b) synchronization
 - c) proximity

Activity 5.4

Answer the questions below.

1. Assuming the CPUs have the same number of cores, which CPU is the fastest?
 - a) CPU_A : 3.5 GHz
 - b) CPU_B : 3.1 GHz
 - c) CPU_C : 2.7 GHz

2. Assuming the CPUs operate at the same clock frequency, which CPU is the fastest?
 - a) CPU_A : dual core
 - b) CPU_B : octa core
 - c) CPU_C : quad core

3. Where is the first place the processor will look for instructions?
 - a) D-cache
 - b) I-cache
 - c) Shared cache

4. Where is the first place the processor will look for data?
 - a) Main memory
 - b) Shared cache
 - c) D-cache

5. If the processor doesn't find data in main memory, where will it look next?
 - a) Storage
 - b) Shared cache
 - c) Register file

Activity 5.5

Answer the questions below.

1. Write the name and specifications of your processor.

Name: Intel Core i5

Cores: Quad-Core

Clock: 3 GHz

2. List, in order, the memory components the processor checks when looking for data. Assume it has already checked the register file.

D-cache, shared cache, main memory, storage

3. What is prefetching, and why does it work?

Prefetching is a process by which the processor retrieves the data it is looking for and then fetches it along with adjacent data items. This is a strategy to improve performance. It is effective because if one data item is needed, there is a high chance that the next adjacent data item will be needed soon as well.

Activity 5.6

Measure your computer's execution time. Type the program below, presented in Python. The program tasks the CPU with adding a long list of numbers and measures the time it takes the CPU to complete the task.

```
import time

def cpu_intensive_task():
    result = 0
    for i in range(1, 1000000):
        result = result + i
    return result

start_time = time.time()
result = cpu_intensive_task()
end_time = time.time()
execution_time = end_time - start_time

print ("Result:", result)
print ("Execution Time:", execution_time, "seconds")
```

Imports the time module.

Defines a function by the specified name. The function adds all the numbers in range between 1 and 1,000,000.

Starts the timer.

Runs the function.

Stops the timer.

Calculates the execution time.

Prints the sum.

Prints the execution time.

1. Run the program for different ranges of numbers and document the execution time for each.

Execution time for adding 1 to 1,000,000 :

Execution time for adding 1 to 10,000,000 :

Execution time for adding 1 to 100,000,000 :
